

“The Adoption and Utilization of Hazard Mitigation Practices by Jurisdictions along Gulf and Atlantic Coasts”

Data processing workflow

Alexander Abuabara, Walter Peacock

abuabara@tamu.edu

2018/Jul (DRAFT)

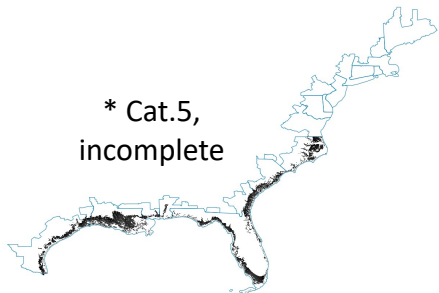
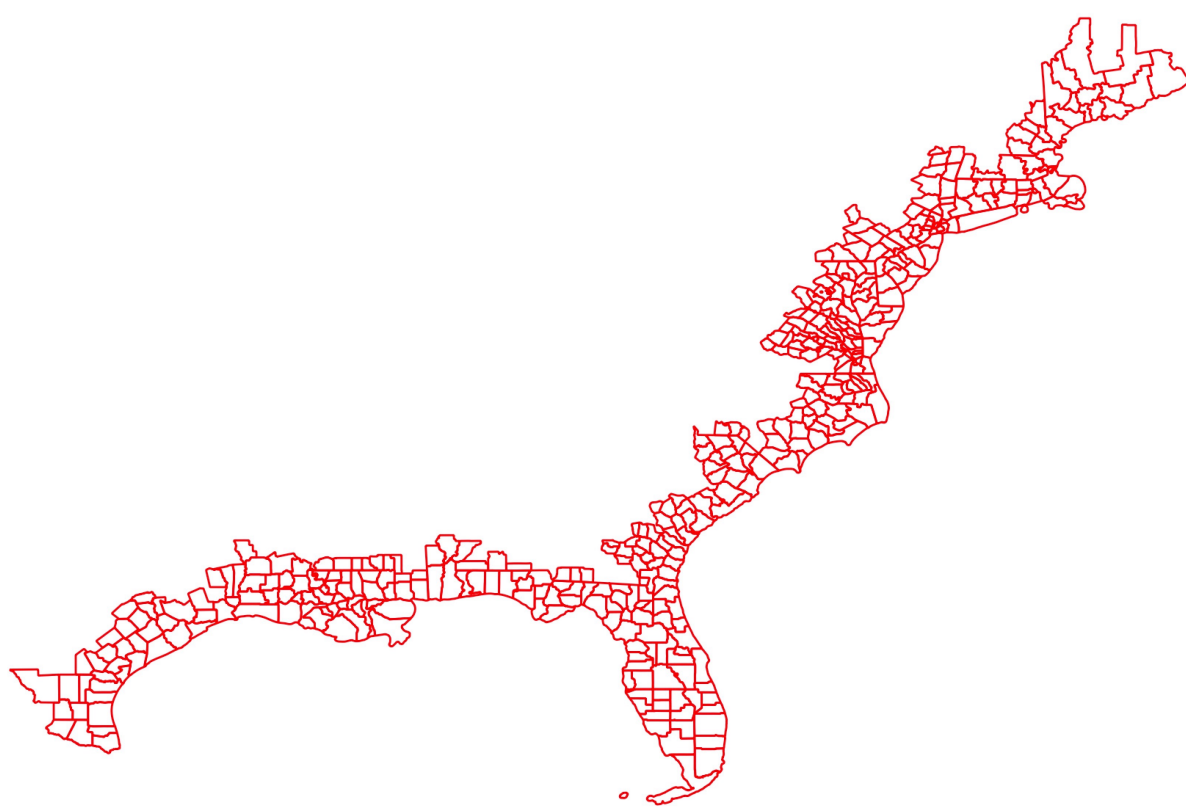
objective

quantify the areas of flood and surge, and the max. windspeed per county and per place (and per county without the places)



A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
	county	type	no_places	no_water	m_file	Size	Source	Last_Update	Area_Total	Area_Water	Area_Levee	Area_100Yr_V	Area_100Yr_Oth	Area_500Yr	Area_Out	Area_Surge_Cat.1	Area_Surge_Cat.2	Area_Surge_Cat.3	Area_Surge_Cat.4	Area_Surge_Cat.5	Max_Wind_cat.I	Max_Wind_cat.II	Max_Wind_cat.III	Max_Wind_cat.IV
1	1003	county	NO	YES	YES	27.3 Mb	NFHL	2/17/16	2029.3	438	0	49.2	273.5	13	1255.4	195.2	251	309	344.9	369	150	161.1	176.2	184.1
2	1003	county	YES	YES	YES	23 Mb	NFHL	2/17/16	1863.1	425.7	0	43	253.2	10.2	1130.8	179.1	222.8	272.7	304.6	326	150	161.1	176.2	184.1
3	1025	county	NO	YES	YES	39.3 Mb	NFHL	7/17/12	1253.6	14.1	0	0	226	0.1	1013	10.9	31.3	49.7	56.6	67	128.3	138.7	147.5	153.1
4	1025	county	YES	YES	YES	38.7 Mb	NFHL	7/17/12	1229	13.9	0	0	223.9	0	990.8	10.9	31.3	49.7	56.6	67	128.3	138.7	147.5	153.1
5	1003	place	NO	YES	YES	345.7 Kb	NFHL	2/17/16	17.3	0.1	0	0	0.5	0.1	16.6	0	0	0	0	0	137	147.7	157.4	164.2
6	1003	place	NO	YES	YES	848.1 Kb	NFHL	2/17/16	17.7	1.2	0	0.5	0.6	0	15.3	0.8	1	1.1	1.3	1.5	143.2	155.6	165.6	173.6
7	1003	place	NO	YES	YES	1 Mb	NFHL	2/17/16	12.1	0	0	0.1	0.4	0	11.5	0.3	0.5	0.8	1	1.1	145.3	158.2	169.1	176.3
8	1003	place	NO	YES	YES	739.8 Kb	NFHL	2/17/16	31.4	0.1	0	0	0.6	0	30.6	0.2	0.6	1.1	1.8	2.4	146.8	157.3	170	175.5
9	1003	place	NO	YES	YES	2.1 Mb	NFHL	2/17/16	31.3	4.9	0	2.9	11	2.5	10	11.3	19.7	23.7	25.5	26.3	149.6	160.4	174.3	181.4
10	1003	place	NO	YES	YES	996.6 Kb	NFHL	2/17/16	16	1.2	0	1	4.9	0.1	8.6	3.7	8.5	12.6	14.3	15.2	147.6	157.8	170.7	176.5
11	1003	place	NO	YES	YES	369.2 Kb	NFHL	2/17/16	5.5	0	0	0	0.4	0	5	0	0	0	0	0	142.9	154.5	165.2	172
12	1003	place	NO	YES	YES	472.4 Kb	NFHL	2/17/16	35	4.7	0	1.6	1.8	0	26.9	3	3.3	3.5	3.8	4.2	142.1	154.2	163.8	171.8
13	1025	place	NO	YES	YES	480.8 Kb	NFHL	7/17/12	15.8	0.2	0	0	1.6	0.1	14	0	0	0	0	0	121.2	131.1	139.4	144.6
14	1025	place	NO	YES	YES	261.5 Kb	NFHL	7/17/12	8.7	0	0	0	0.5	0	8.2	0	0	0	0	0	114.3	121.9	132.1	136.1

Adoption background



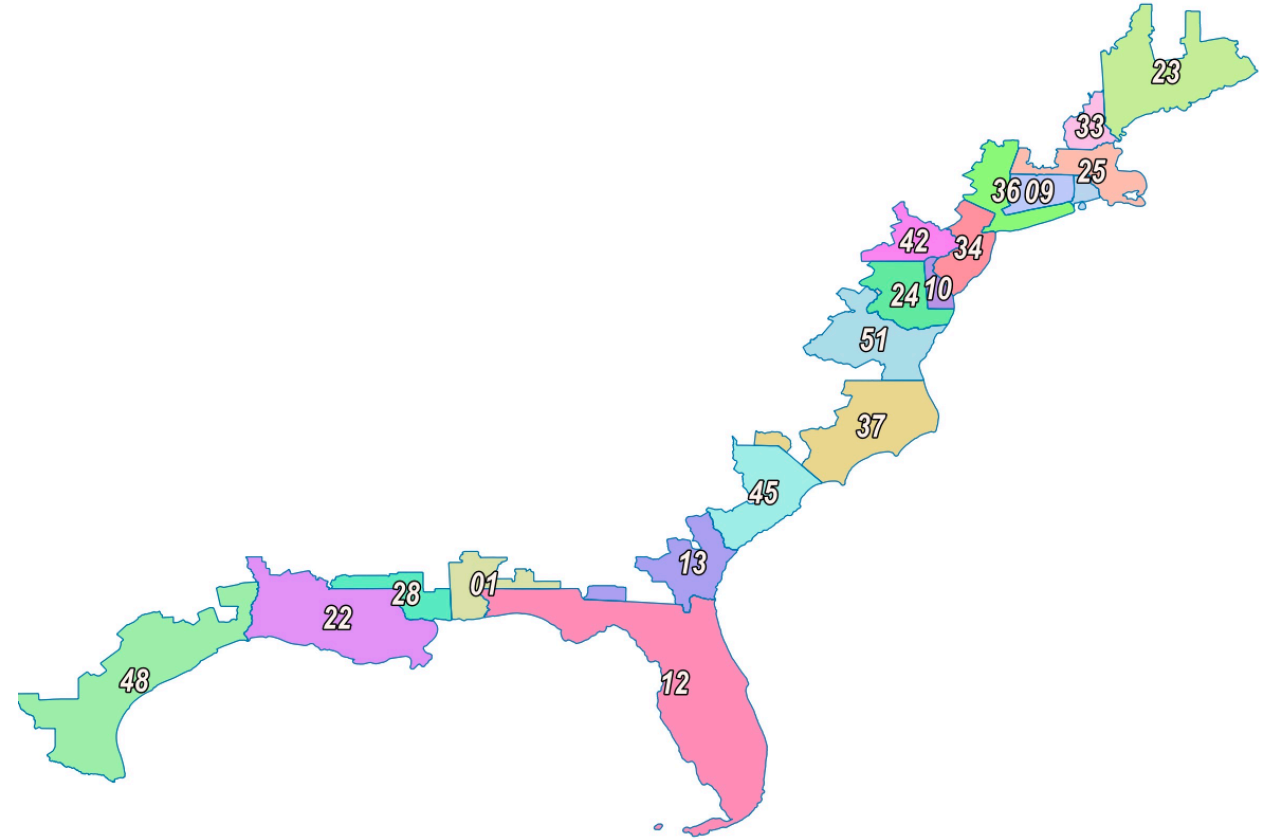
Wind:

- ✓ ACSE7-16 **cat.I, II, III, IV**
cat. refers to the use or occupancy of buildings and structures

Surge:

- ✓ SLOSH MOMs HighTide,
cat.1 to 5*
cat. refers to hurricane

20 states
438 counties



Flood [counties]:

- ✓ 380 NFHL
- ✓ 49 Q3 (<2005)
- ✓ 9 no data

few facts

* according census 2010

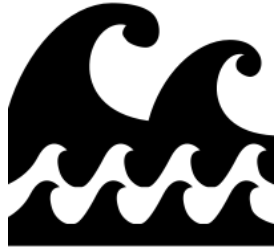
total number of census counties processed	429
total area processed	678,664,831,302 m2 average 1,581,969,304 m2 / county 7% of the USA area *total USA area: 9,826,675,000,000 m2 **total Texas area: 695,621,130,000 m2
total of occupied households covered	34,549,835 units
average of occupied households	80,536 units per county
total population in occupied households	89,005,490 people (28.8 % of the US pop.)
average population in occupied households	207,472 people per county

few facts after data process

total number of census counties	429
total size of flood files processed	25,859 Mb = 25 Gb
total area processed	678,664,831,302 m2 average 1,581,969,304 m2 / county
area of water	13,287,366,058 m2 1.9 %
total under 100Yr	167,631,409,126 m2 24.7%
total under 500Yr	12,818,178,627 m2 1.9%
total under Levee protection	2,015,954,601 m2 0.3%
total out of flood risk	482,911,922,890 m2 71.2%



Flooding



Surge



Wind



Policies



Project properties

- Ellipsoid: **Clarke 1866 Authalic Sphere**
 - Units for distance meas.: **Meters**
 - Units for area measurement: **Square meters**
 - Projection: **EPSG:2163 US National Atlas Equal Area**
- Information source:
- US Geological Survey
 - Western Geographic Science Center





Floodplain

base year for **boundaries** &
water features: **Census Tiger files 2016**

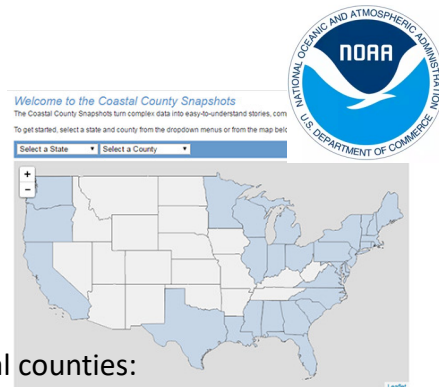
flood data collection



*Mitigation Full data and
sample 6-23-16 no
identifiers.dta*

Michelle's

- 438 Atlantic counties
- 3,531 urban places



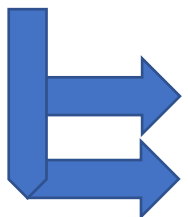
NOAA coastal counties:

- 246 NOAA counties not in the research
- 427 NOAA matches
- 11 counties in the research are not in the NOAA database

**United States™
Census**
Bureau

Census counties:

- 438 out of 3,233 US counties
- 6,947 urban places



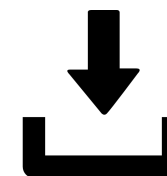
2015 TIGER shapefiles

- counties (and equivalent)
- water



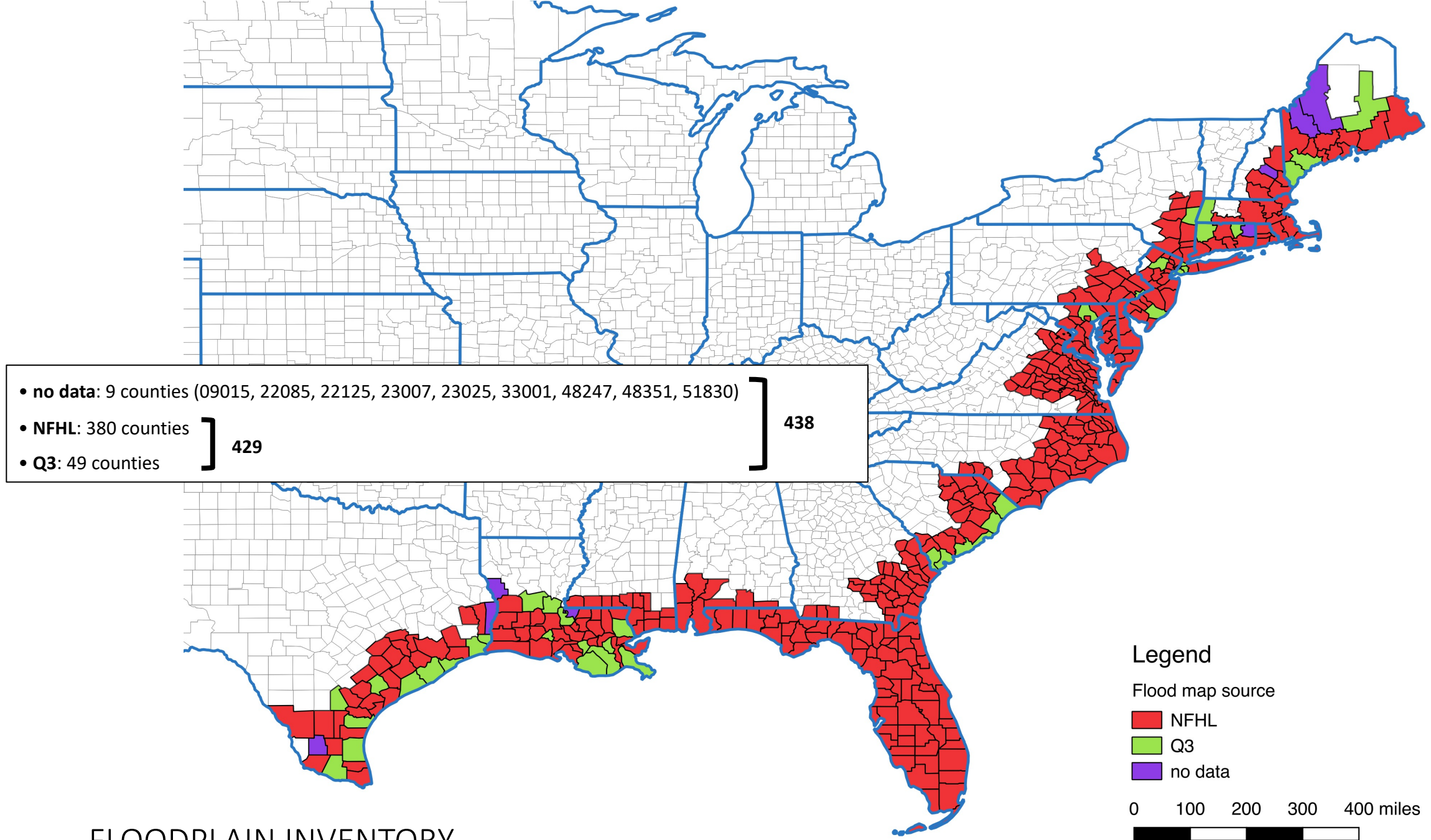
Q3: 49 counties

- countrywide: CD-ROM borrowed from Michigan's library geocomm data
- for Texas: tnris.org
- missing ones: bought from data.geocomm.com
(7 counties: PA: Philadelphia, and SC: Beaufort, Charleston, Colleton, Darlington, Dorchester, Georgetown, Horry, Jasper)



NFHL: 380 counties

- countrywide: FEMA
www.floodmaps.fema.gov/NFHL/status.shtml
(my last update 2018/Jul)

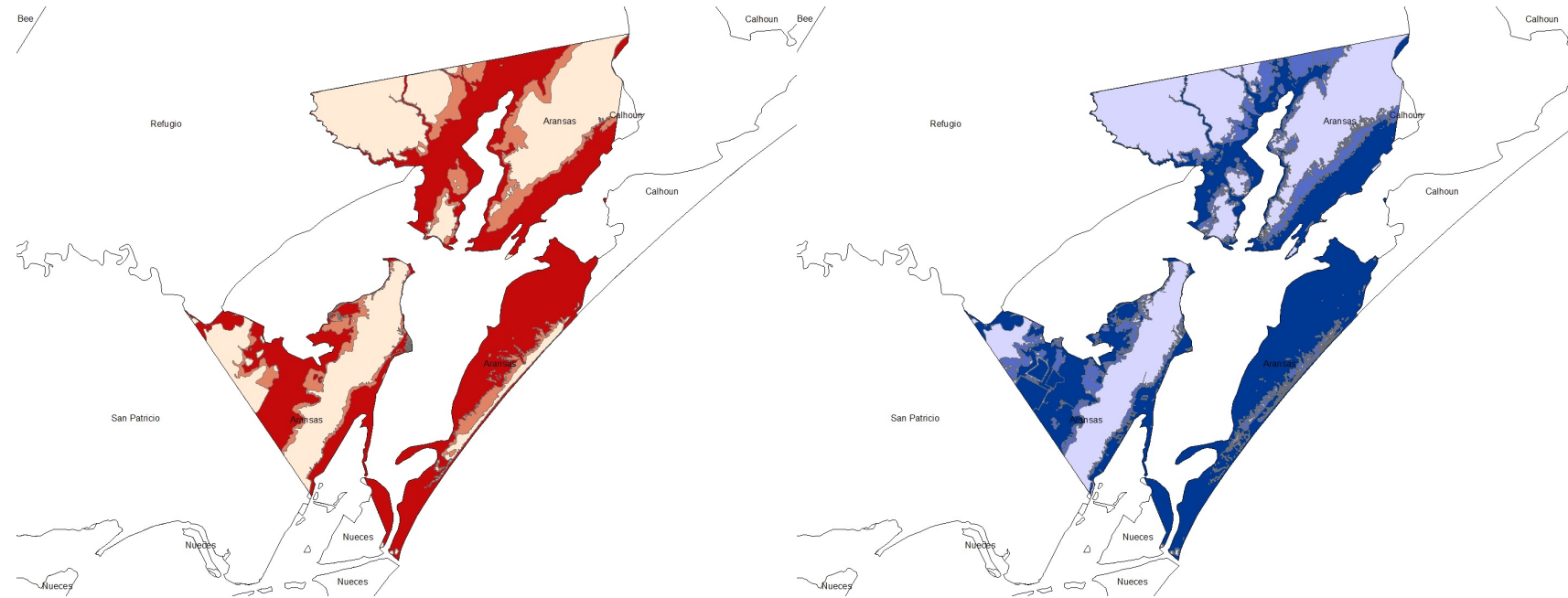


FLOODPLAIN INVENTORY

limitations

- Q3 and NFHL can have differences, in resolution and classification

- Best Q3 resolution 1:24.000
- NFHL can be as good as 1:500



- 9 counties in the research area still do not have any kind of flooding information:
 - Windham, CT
 - West Feliciana & Sabine, LA
 - Franklin & Somerset, ME
 - Belknap, NH
 - Dorchester, SC
 - Jim Hogg & Newton, TX

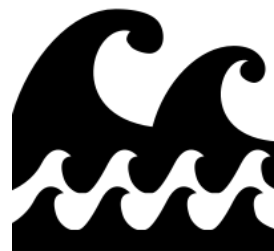
floodplain reclassification

New_Code	Re-classification for Q3	Re-classification for NFHL
100Yr	V, VE, A, A1-A30, A99, AE, AH, AO, AR	VE, A, AE, AH, AO
500Yr	B, X500, X (Shaded)	X (0.2 pct annual change flood hazard)
Levee	X (reduced flood risk due to levee)	—
Out	C, D, X, X (Unshaded), UNDES	Area Not Included, X (minimal flood hazard)

Possible to find a lot of sources referring ultimately to FEMA ... such as:

NFHL: <http://www.floodmaps.com/zones.htm>

Q3: <http://www.mass.gov/anf/docs/itd/services/massgis/q3floodzonescodetable.pdf>



surge

surge data collection



National Storm Surge Hazard Maps - Version 2



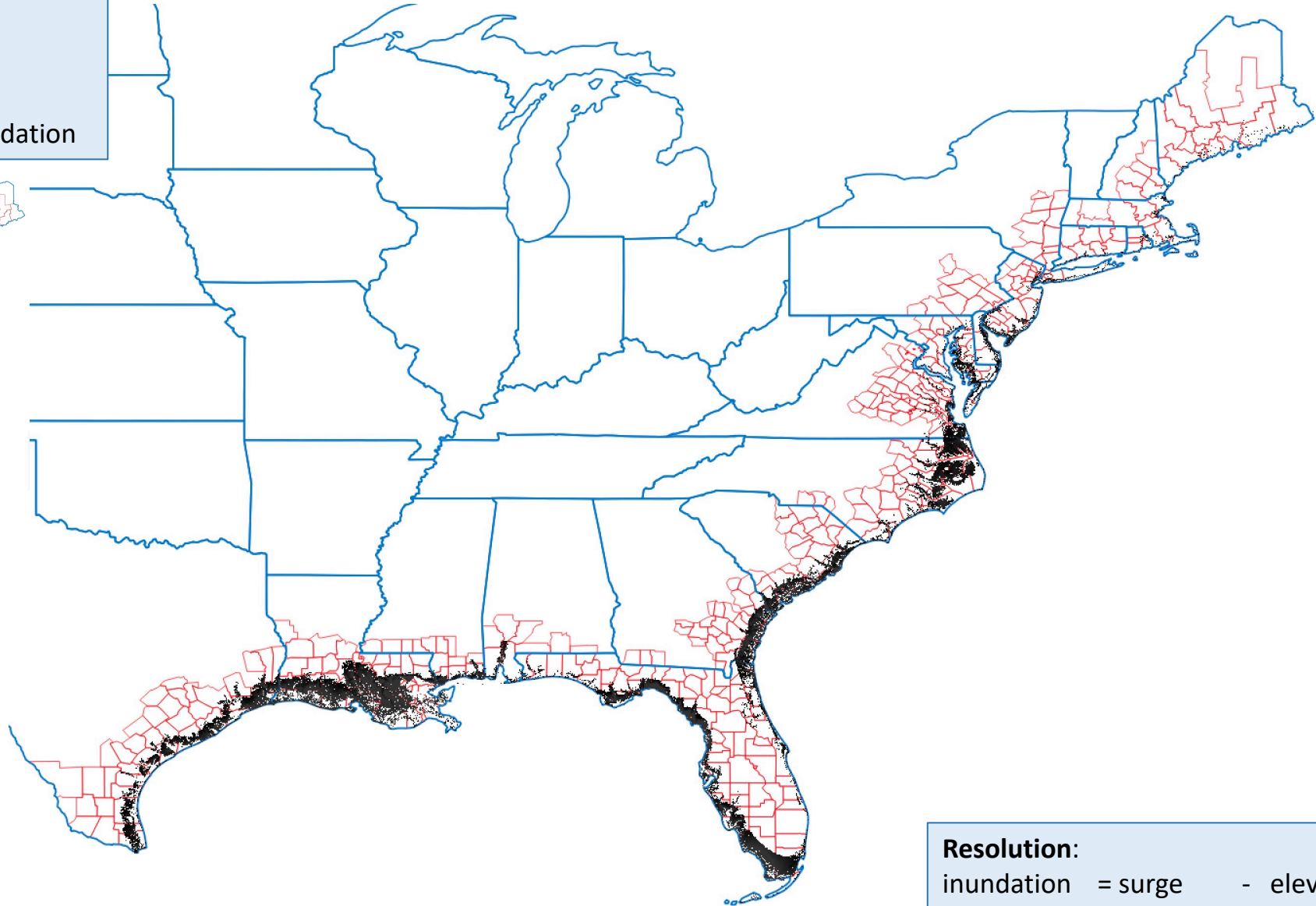
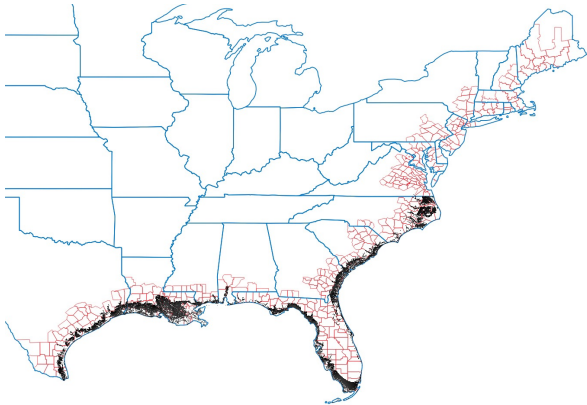
- updated topography data (DEM)
- utilizes the hydrodynamic Sea, Lake, and Overland Surges from Hurricanes (SLOSH) model to simulate storm surge from tropical cyclones
- surge composites – Maximum Envelopes of Water (MEOWs) and Maximum of MEOWs (MOMs)
- storm surge inundation datasets are created using the **high tide** scenario **SLOSH MOM** products for all regions
- GeoTIFFs are 8-bit integer raster datasets that correspond to 1 ft inundation bins (e.g., **Class Value 1 corresponds to the 0-1 ft inundation bin, Class Value 2 corresponds to the 1-2 ft inundation bin**, and so on)
- **the maximum Class Value is 21**, and inundation in excess of 20 ft is assigned a Class Value of 21. A
- **class Value of 99 is assigned to leveed areas**

Update schedule: The storm surge hazard maps and GIS data (if a new version is available) will be updated around the beginning of June each year.

inundation overview

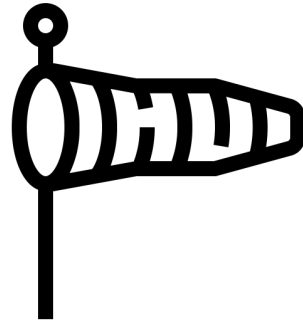
Inundation:

- CAT1 to CAT5
(CAT5 do not cover all the research area)
- MOMs (High Tide)
- calculated the areas with ANY inundation



Resolution:

inundation	= surge	- elevation
(~30x30 m)	(1x1 km)	(10x10 m)



(*new*) wind fields

wind data collection

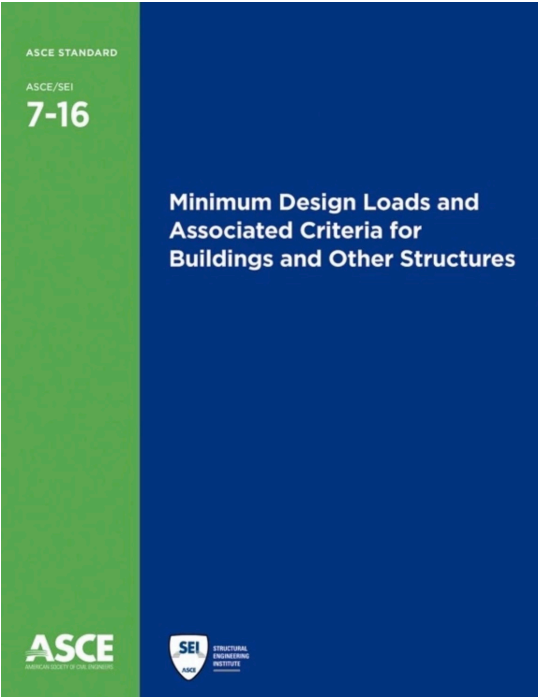
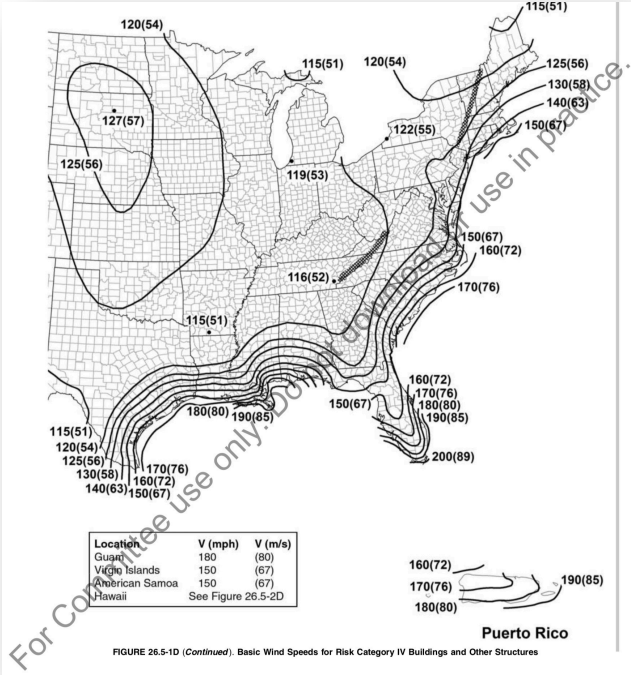
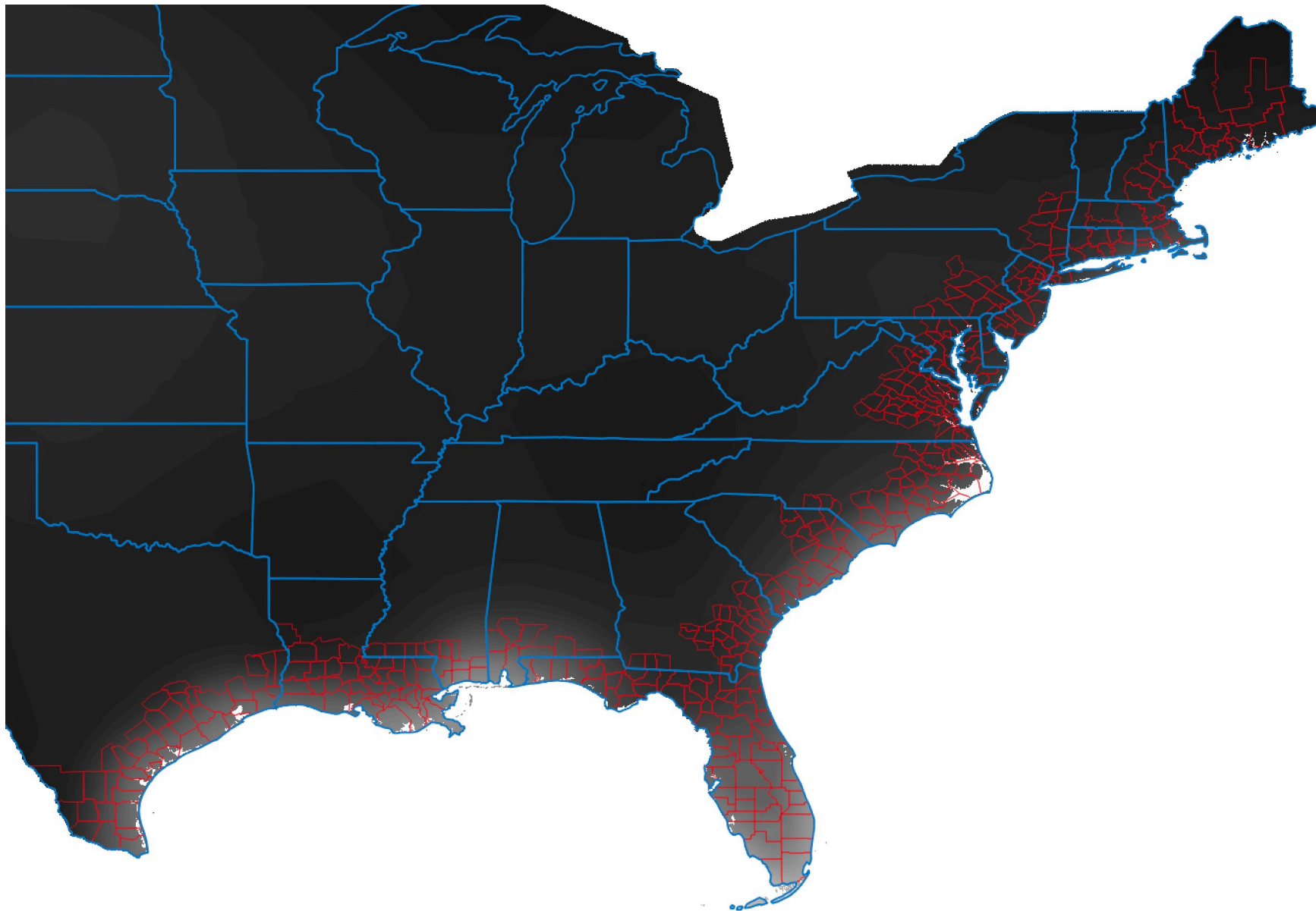


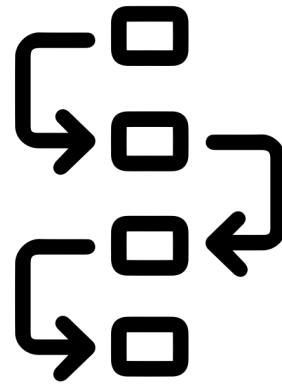
Table 1.5-1 Risk Category of Buildings and Other Structures for Flood, Wind, Snow, Earthquake, and Ice Loads

Use or Occupancy of Buildings and Structures	Risk Category
Buildings and other structures that represent low risk to human life in the event of failure	I
All buildings and other structures except those listed in Risk Categories I, III, and IV	II
Buildings and other structures, the failure of which could pose a substantial risk to human life	III
Buildings and other structures, not included in Risk Category IV, with potential to cause a substantial economic impact and/or mass disruption of day-to-day civilian life in the event of failure	
Buildings and other structures not included in Risk Category IV (including, but not limited to, facilities that manufacture, process, handle, store, use, or dispose of such substances as hazardous fuels, hazardous chemicals, hazardous waste, or explosives) containing toxic or explosive substances where the quantity of the material exceeds a threshold quantity established by the Authority Having Jurisdiction and is sufficient to pose a threat to the public if released ^a	
Buildings and other structures designated as essential facilities	IV
Buildings and other structures, the failure of which could pose a substantial hazard to the community	
Buildings and other structures (including, but not limited to, facilities that manufacture, process, handle, store, use, or dispose of such substances as hazardous fuels, hazardous chemicals, or hazardous waste) containing sufficient quantities of highly toxic substances where the quantity of the material exceeds a threshold quantity established by the Authority Having Jurisdiction and is sufficient to pose a threat to the public if released ^a	
Buildings and other structures required to maintain the functionality of other Risk Category IV structures	



wind overview





Adoption workflow



Workflow outline

(approx. 1000 lines)

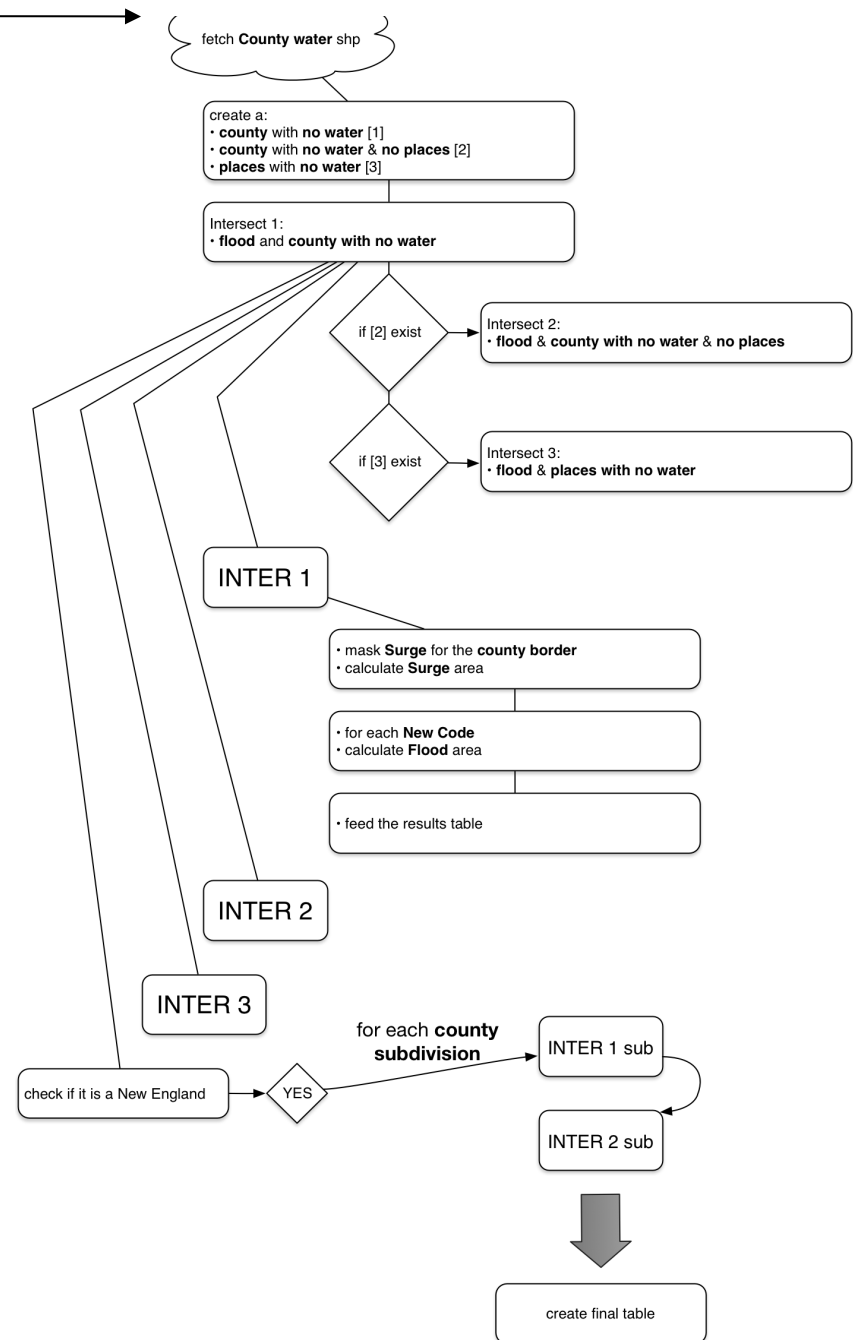
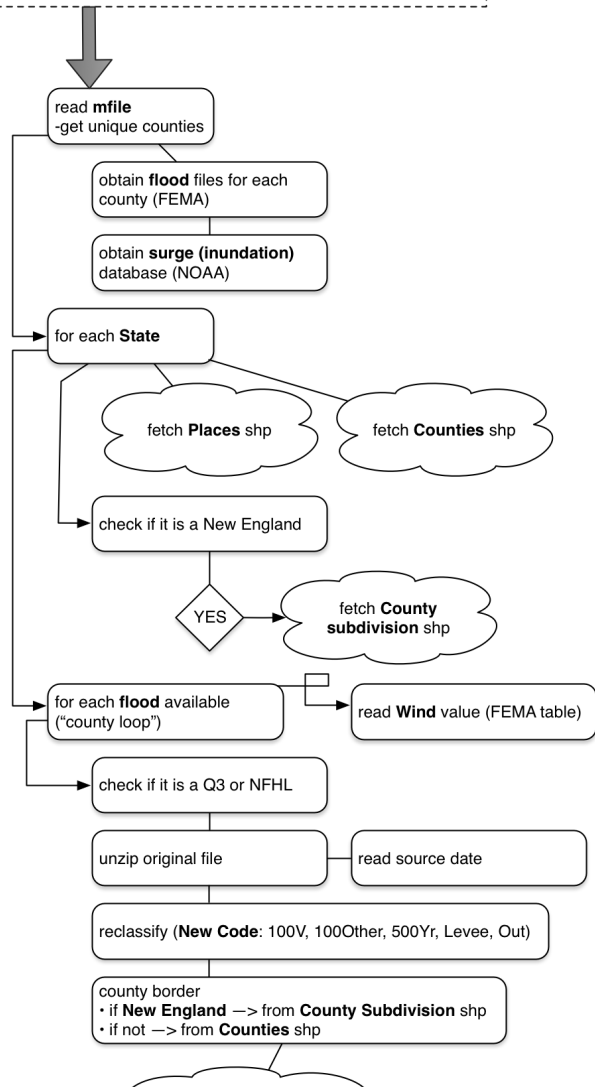


- m_file:
 - m_file: counties
 - m_file: places
- intersections:
 - Inter1 = County – water x Floodplain
 - Inter2 = County – water – places x Floodplain
 - Inter3 = Places – water x Floodplain
- areas:
 - for each **Inter1**, for **Inter2**, for **Inter3**
 - Flood
 - Wind
 - Surge

Non-destructive → all data just reclassified and quantified

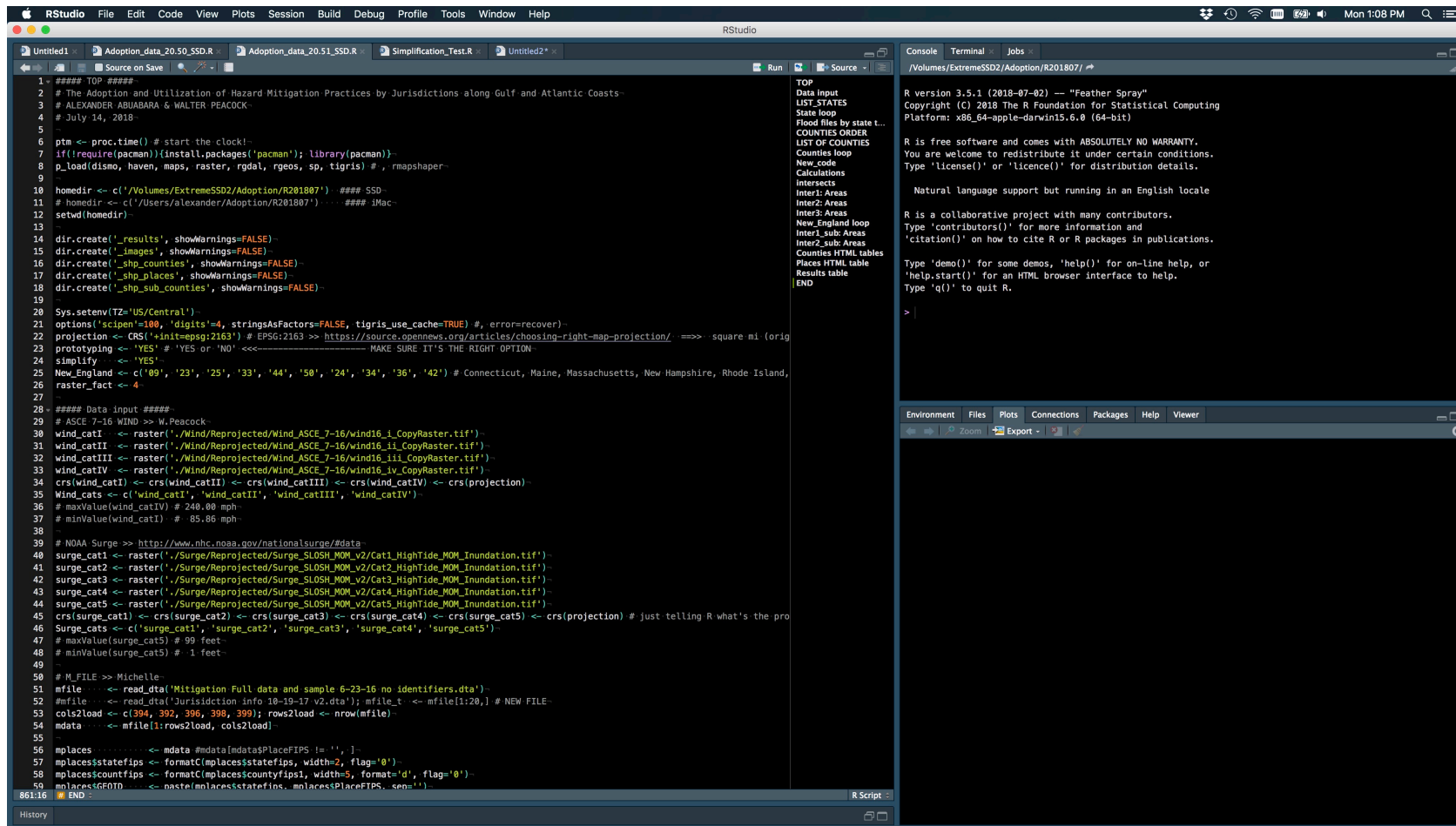
INITIAL SETUP
Projection —> EPSG:2163 US National Atlas Equal Area
Packages —> haven, tigris, rgdal, rgeos, sp, raster
New England States —> 09, 23, 25, 33, 44, 50, 24, 34, 36, 42

for every SHP:
1) project
2) buffer with 0 width



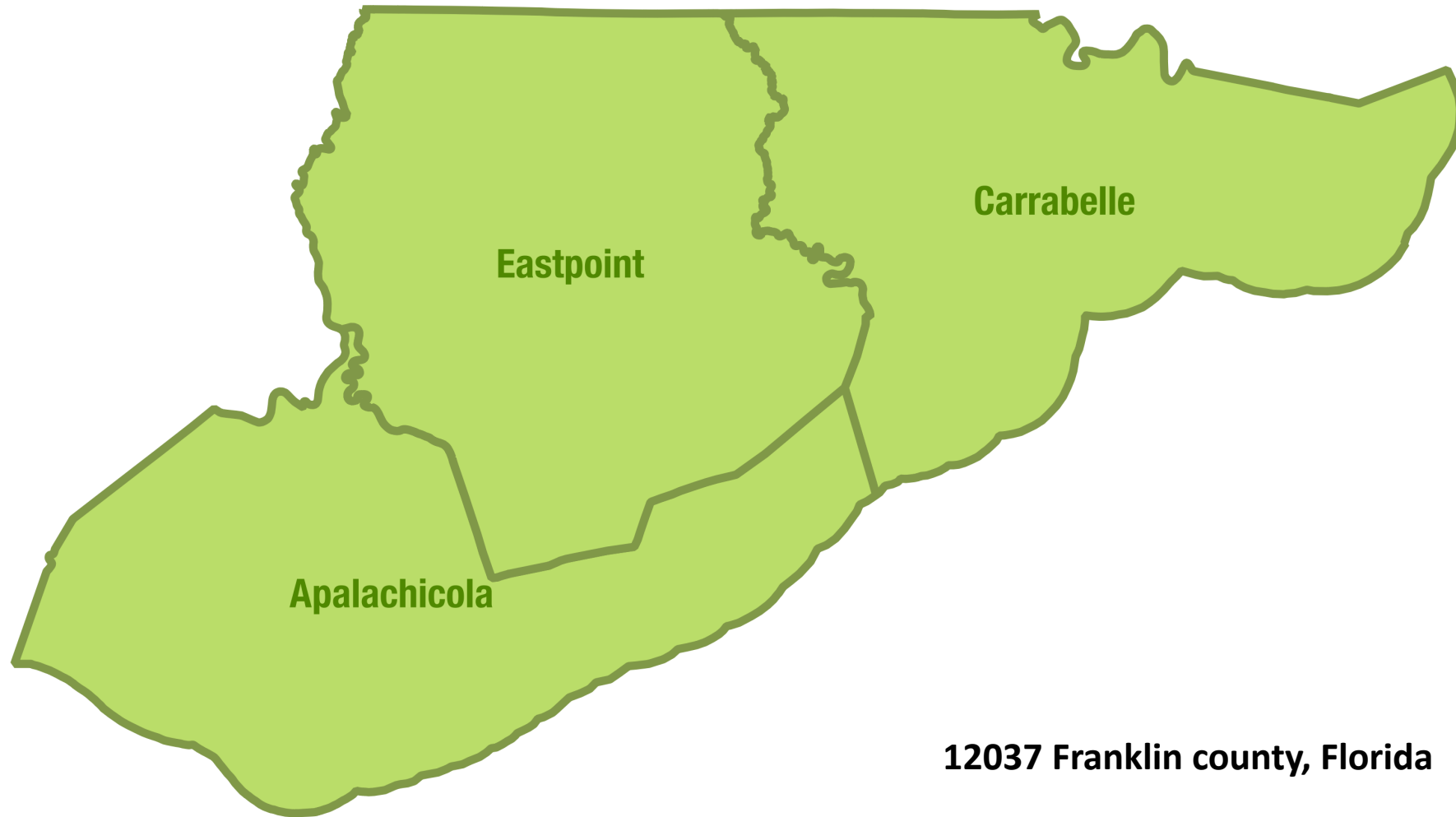
Summary

- Runs on Windows, Mac, Linux (including the TAMU supercomputing)
- All counties takes: 161,914 sec ~ approx. 45 hours of computing time (estimated time based on the last run I did)
- Requires R (Rstudio recommended), and the following packages: *haven*, *maps*, *raster*, *rgdal*, *rgeos*, *sp*, *tigris*

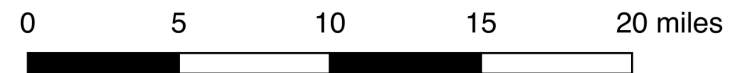


```
1 ##### TOP #####
2 # The Adoption and Utilization of Hazard Mitigation Practices by Jurisdictions along Gulf and Atlantic Coasts-
3 # ALEXANDER ABUABARA & WALTER PEACOCK-
4 # July 14, 2018
5
6 ptm <- proc.time() # start the clock!-
7 if(!require(pacman)){install.packages('pacman'); library(pacman)}-
8 p_load(dismo, haven, maps, raster, rgdal, rgeos, sp, tigris) # , rmapshaper-
9
10 homedir <- c('/Volumes/ExtremeSSD2/Adoption/R201807') ##### SSD-
11 # homedir <- c('/Users/alexander/Adoption/R201807') ##### iMac-
12 setwd(homedir)-
13
14 dir.create('results', showWarnings=FALSE)-
15 dir.create('images', showWarnings=FALSE)-
16 dir.create('shp_counties', showWarnings=FALSE)-
17 dir.create('shp_places', showWarnings=FALSE)-
18 dir.create('shp_sub_counties', showWarnings=FALSE)-
19
20 Sys.setenv(TZ='US/Central')-
21 options('scipen=100', 'digits=4', stringsAsFactors=FALSE, tigris_use_cache=TRUE) #, error=recover)-
22 projection <- CRS('+init=epsg:2163') # EPSG:2163 >> https://source.opennews.org/articles/choosing-right-map-projection/ ==> square mi (orig
23 prototyping <- 'YES' # 'YES' or 'NO' <<< MAKE SURE IT'S THE RIGHT OPTION-
24 simplify <- 'YES'-
25 New_England <- c('09', '23', '25', '33', '44', '50', '24', '34', '36', '42') # Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island,
26 raster_fact <- 4-
27
28 ##### Data input #####
29 # ASCE 7-16 WIND >> W.Peacock-
30 wind_catI <- raster('/Wind/Reprojected/Wind_ASCE_7-16/wind16_i_CopyRaster.tif')-
31 wind_catII <- raster('/Wind/Reprojected/Wind_ASCE_7-16/wind16_ii_CopyRaster.tif')-
32 wind_catIII <- raster('/Wind/Reprojected/Wind_ASCE_7-16/wind16_iii_CopyRaster.tif')-
33 wind_catIV <- raster('/Wind/Reprojected/Wind_ASCE_7-16/wind16_iv_CopyRaster.tif')-
34 crs(wind_catI) <- crs(wind_catII) <- crs(wind_catIII) <- crs(wind_catIV) <- crs(projection)-
35 wind_cats <- c('wind_catI', 'wind_catII', 'wind_catIII', 'wind_catIV')-
36 # maxValue(wind_catIV) # 240.00 mph-
37 # minVal(wind_catI) # 85.86 mph-
38
39 # NOAA Surge >> http://www.nhc.noaa.gov/nationalsurge/#data-
40 surge_cat1 <- raster('/Surge/Reprojected/Surge_SLOSH_MOM_v2/Cat1_HighTide_MOM_Inundation.tif')-
41 surge_cat2 <- raster('/Surge/Reprojected/Surge_SLOSH_MOM_v2/Cat2_HighTide_MOM_Inundation.tif')-
42 surge_cat3 <- raster('/Surge/Reprojected/Surge_SLOSH_MOM_v2/Cat3_HighTide_MOM_Inundation.tif')-
43 surge_cat4 <- raster('/Surge/Reprojected/Surge_SLOSH_MOM_v2/Cat4_HighTide_MOM_Inundation.tif')-
44 surge_cats <- raster('/Surge/Reprojected/Surge_SLOSH_MOM_v2/Cat1_HighTide_MOM_Inundation.tif')-
45 crs(surge_cat1) <- crs(surge_cat2) <- crs(surge_cat3) <- crs(surge_cat4) <- crs(projection) # just telling R what's the pro
46 Surge_cats <- c('surge_cat1', 'surge_cat2', 'surge_cat3', 'surge_cat4', 'surge_cats')-
47 # maxValue(surge_cats) # 99 feet-
48 # minVal(surge_cats) # 1 feet-
49
50 # R_FILE >> Michelle-
51 mfile <- read_dta('Mitigation_Full_data_and_sample_6-23-16_no_identifiers.dta')-
52 mfile <- read_dta('Jurisdiction_Info_10-19-17_v2.dta'); mfile_t <- mfile[1:20,] # NEW FILE-
53 cols2load <- c(304, 302, 306, 398, 399); rows2load <- nrow(mfile)-
54 mdata <- mfile[1:rows2load, cols2load]-
55
56 mplaces <- mdata #mdata[mdata$PlaceFIPS != '', ]-
57 mplaces$statefips <- formatC(mplaces$statefips, width=2, flag='0')-
58 mplaces$countfips <- formatC(mplaces$countfips, width=5, format='d', flag='0')-
59 mplaces$GEOID <- paste(mplaces$statefips, mplaces$PlaceFIPS, sep='')-
60
61 ##### END #####
```

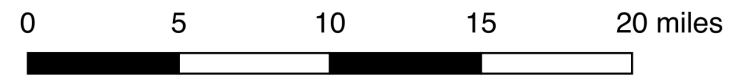
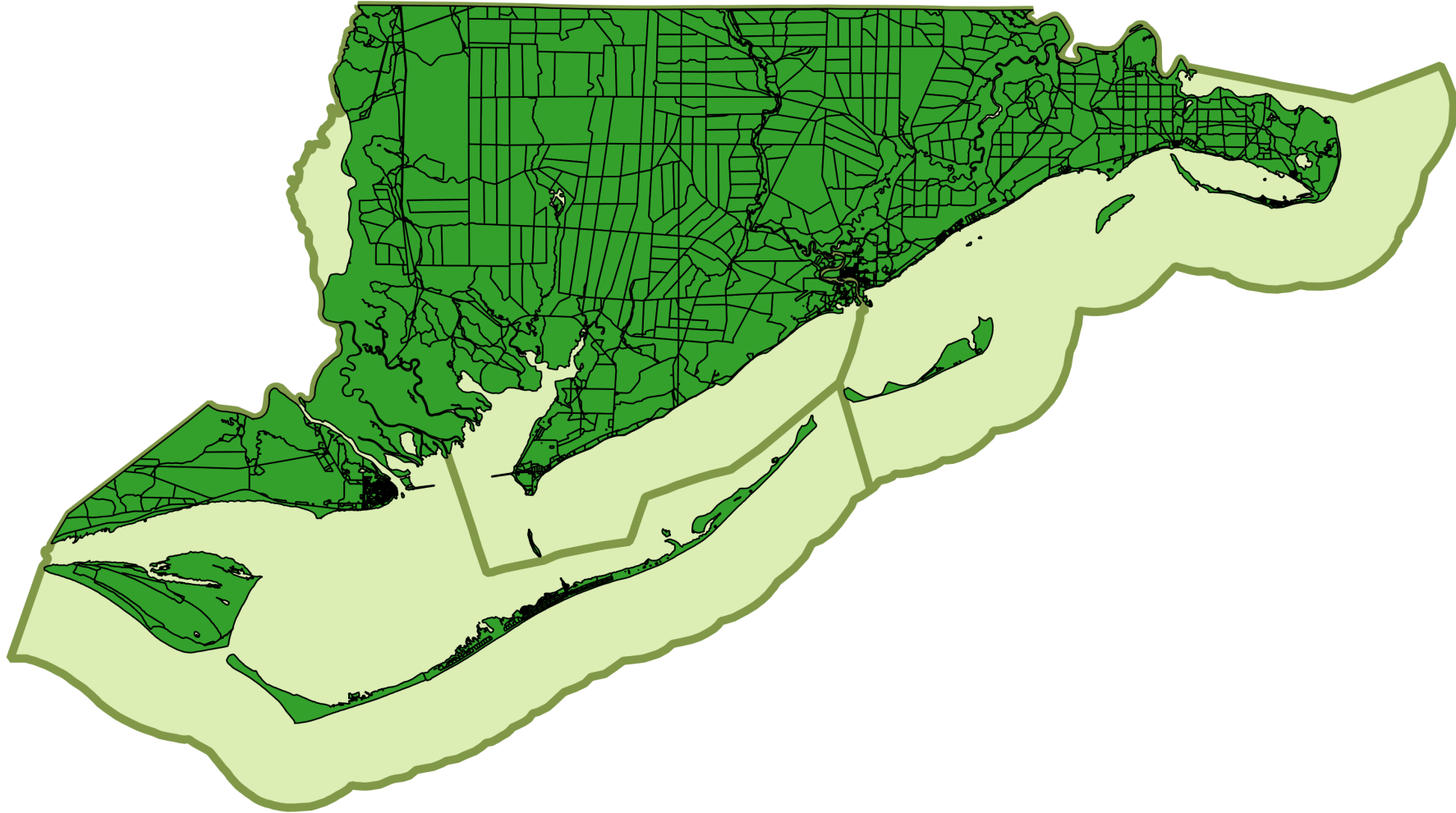
county boundaries
including its subdivisions for the New England states



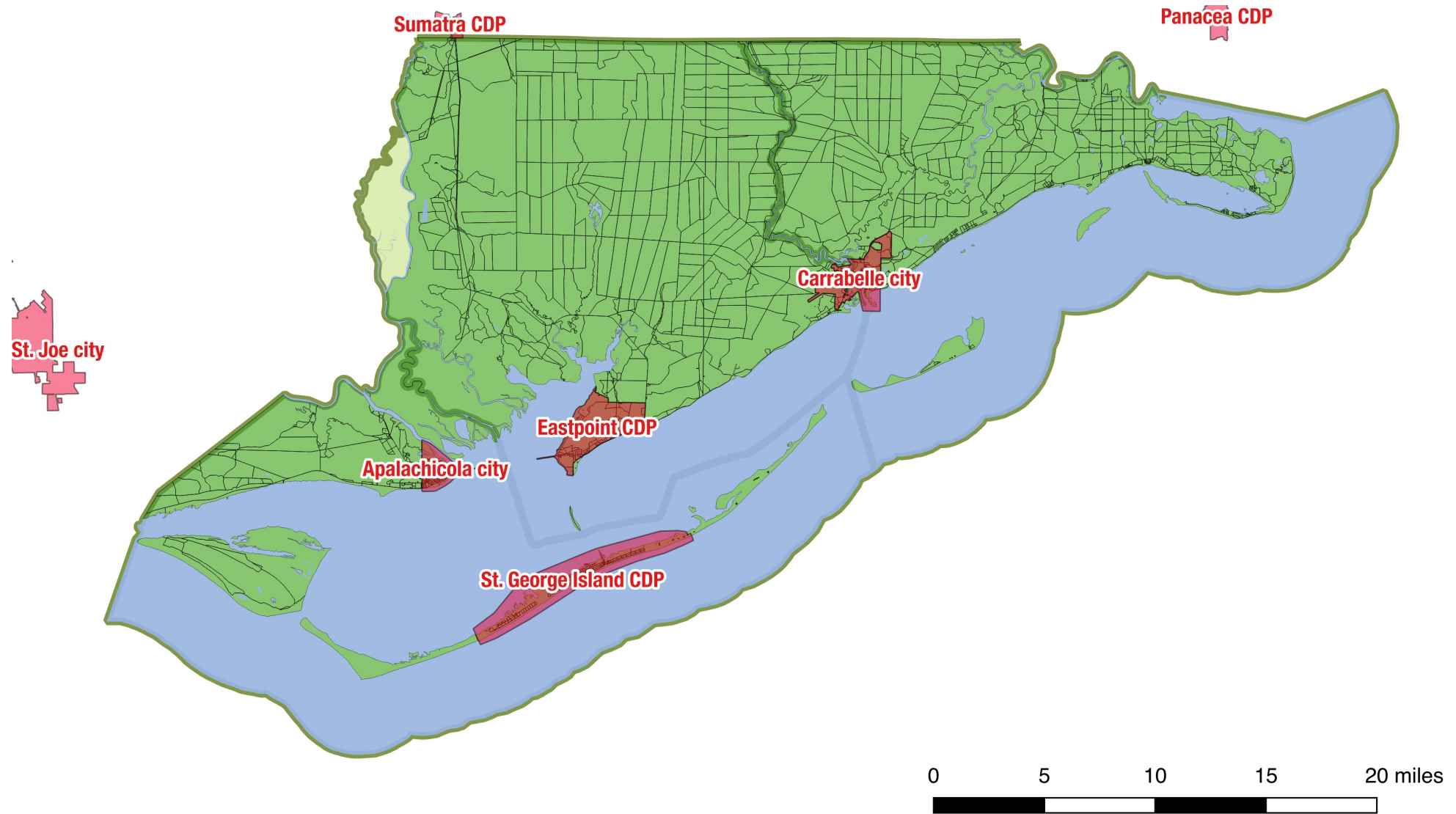
12037 Franklin county, Florida



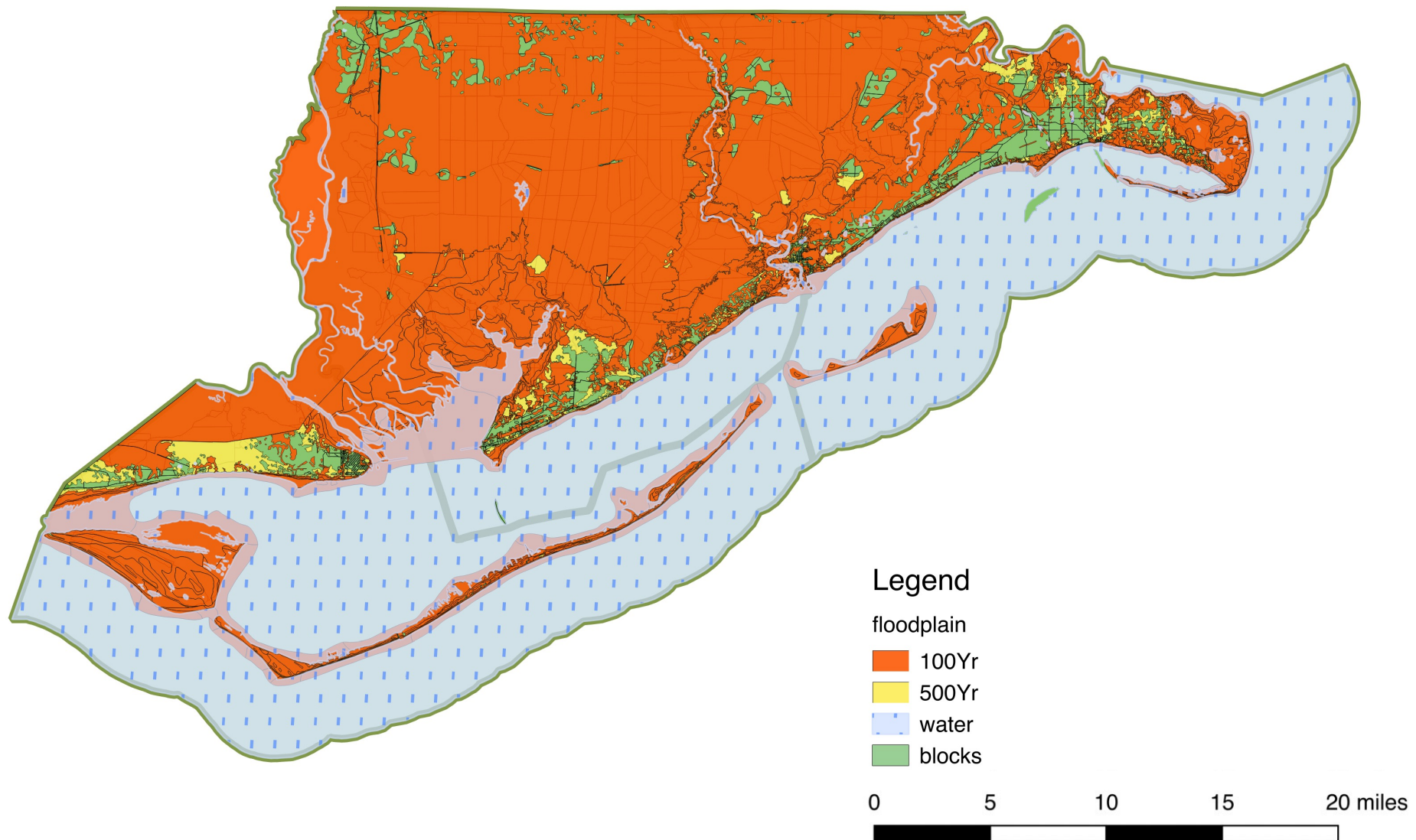
area of land > 0



identify water & places

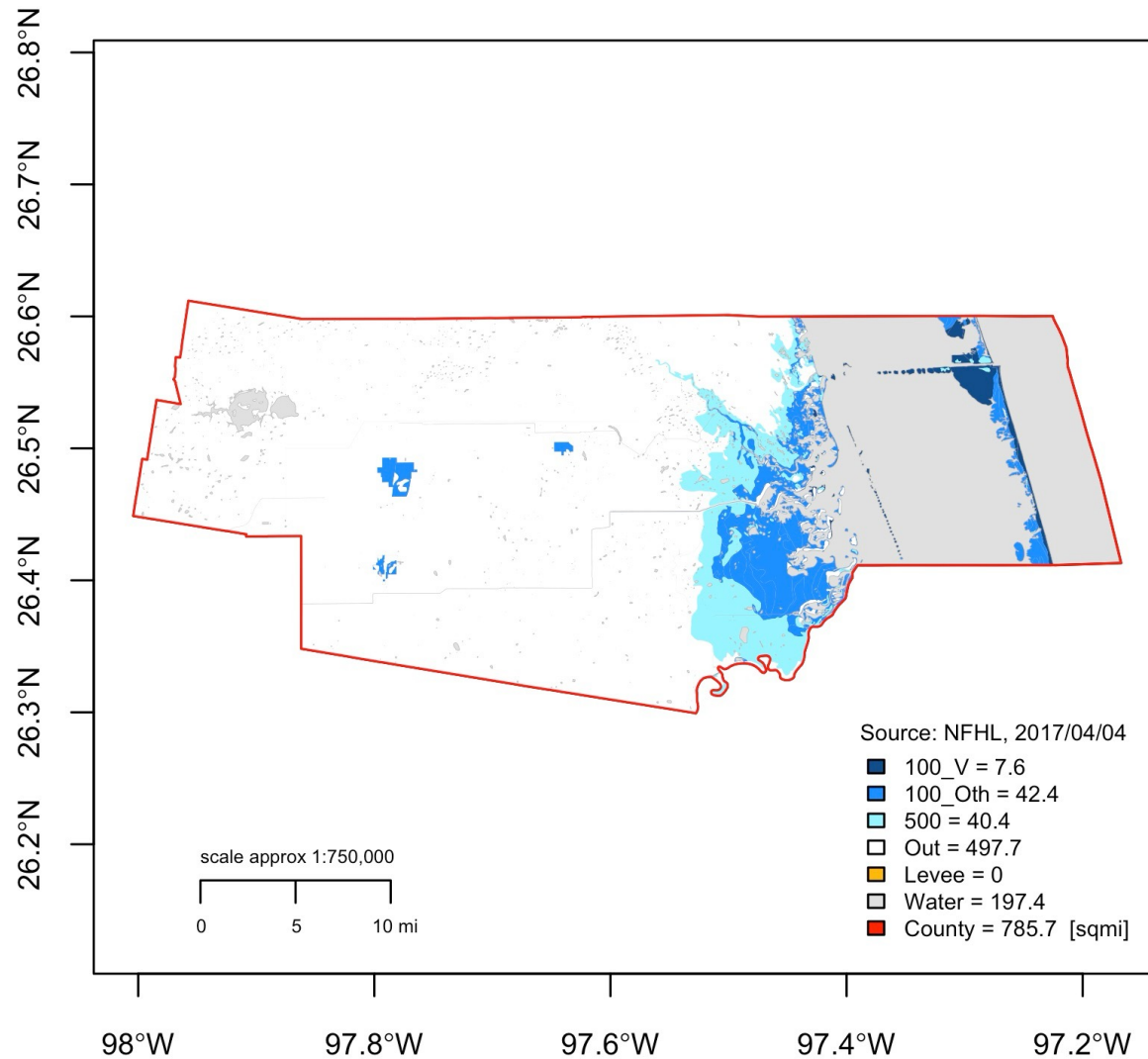


floodplain, reclassification and tidy
original coverage

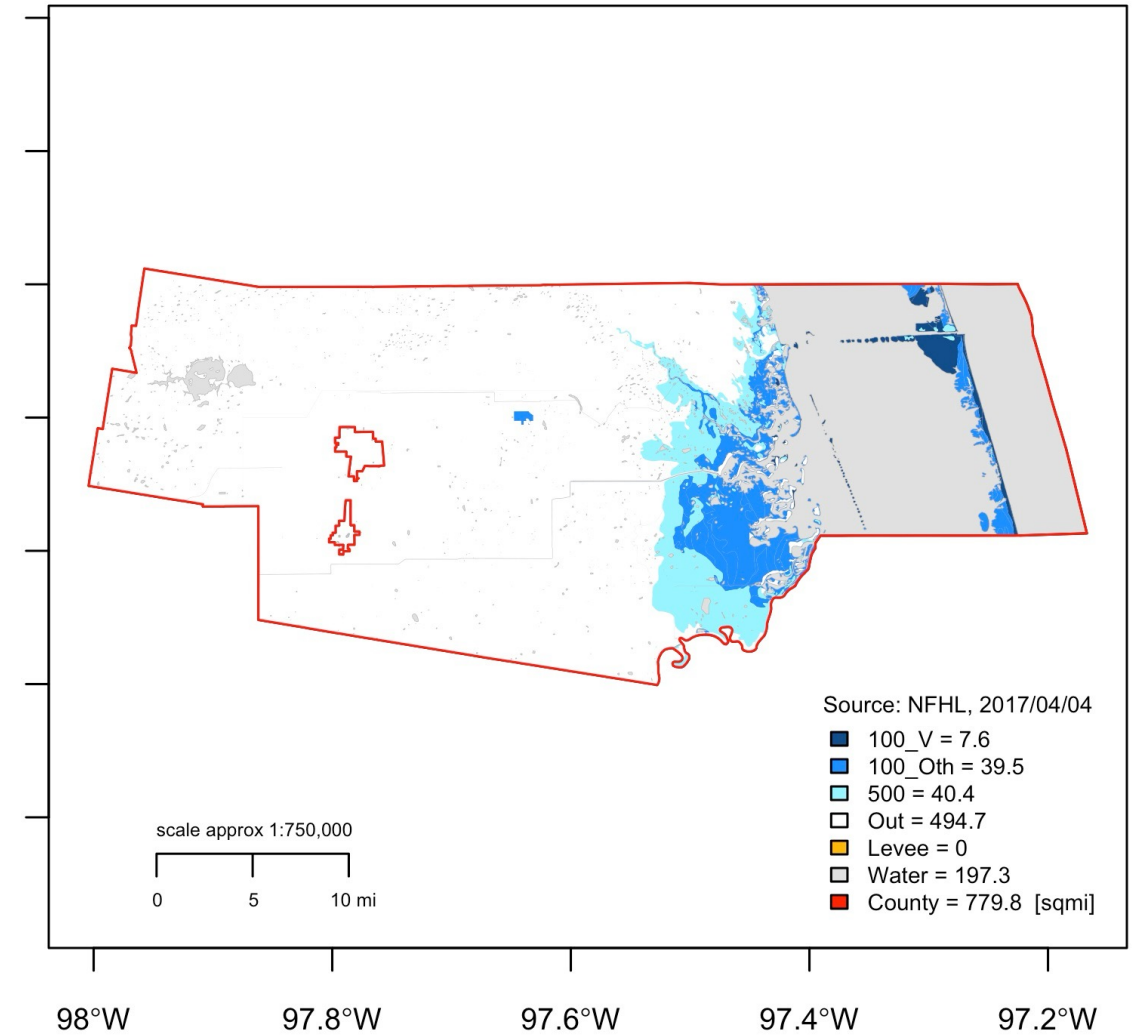


Example: WILLACY, TX

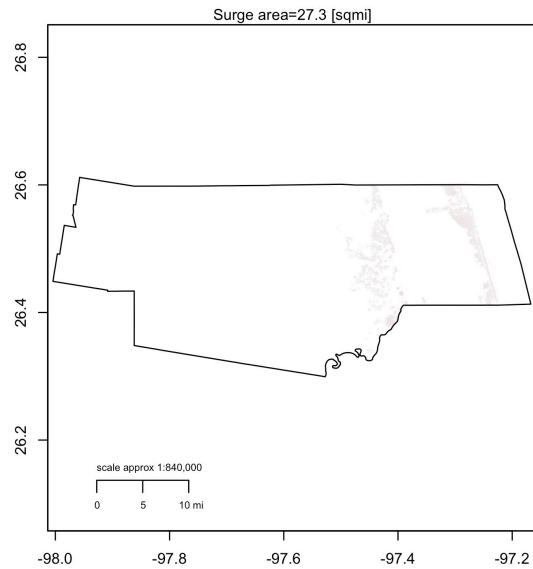
Floodplain, county 48489



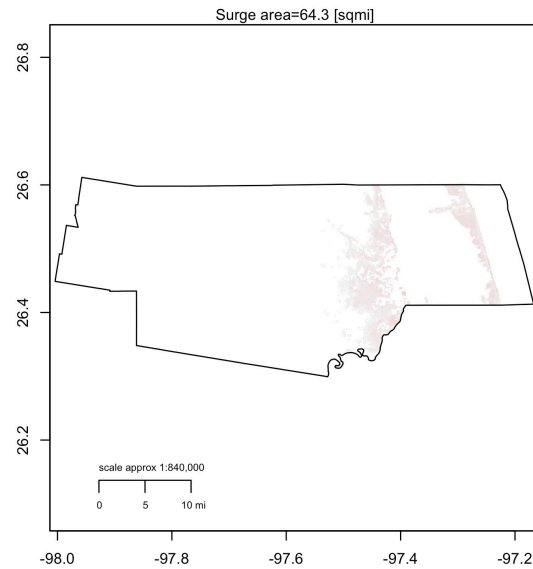
Floodplain, county 48489 (no places)



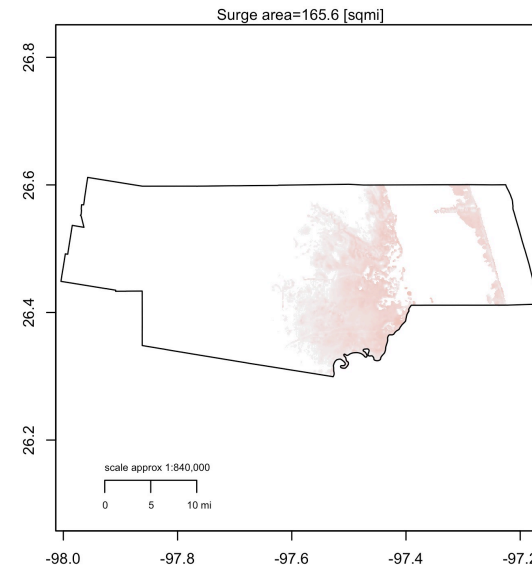
Surge cat. 1, county 48489



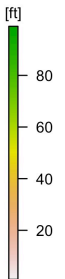
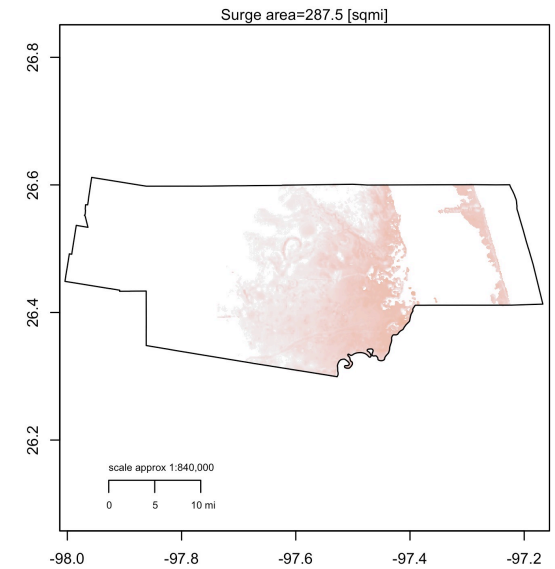
Surge cat. 2, county 48489



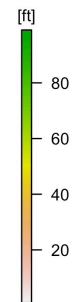
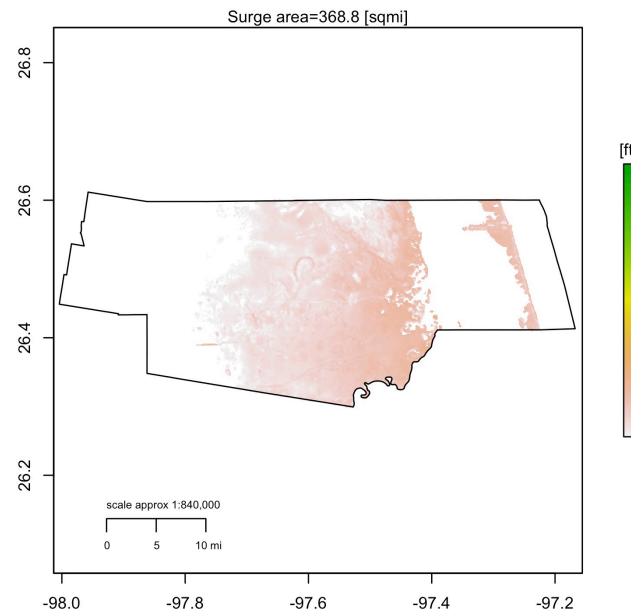
Surge cat. 3, county 48489



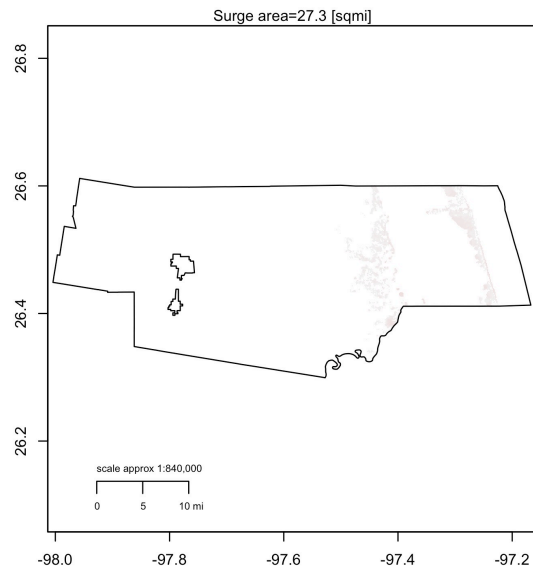
Surge cat. 4, county 48489



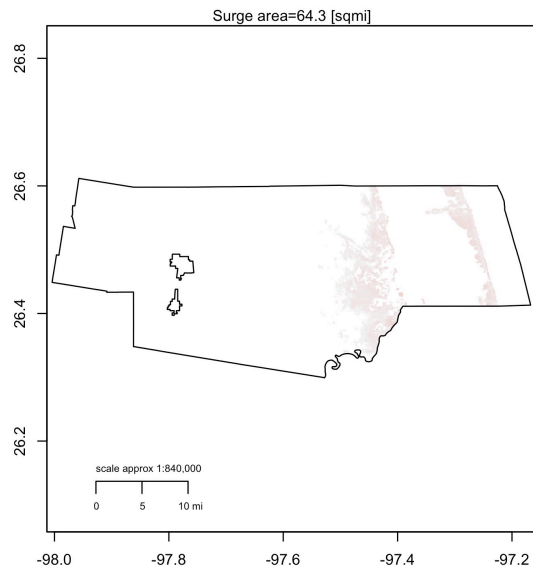
Surge cat. 5, county 48489



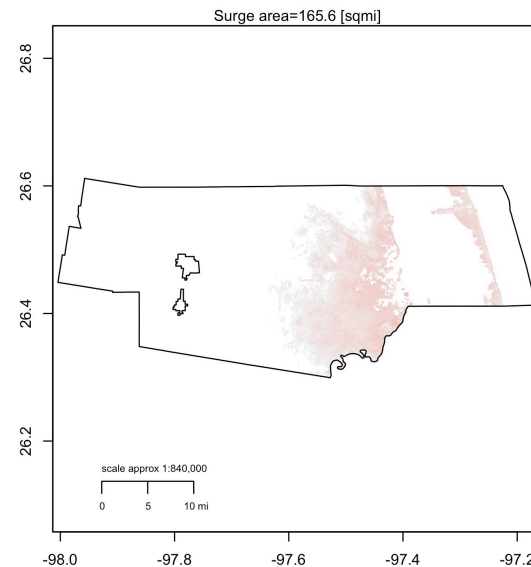
Surge cat. 1, county 48489 (no places)



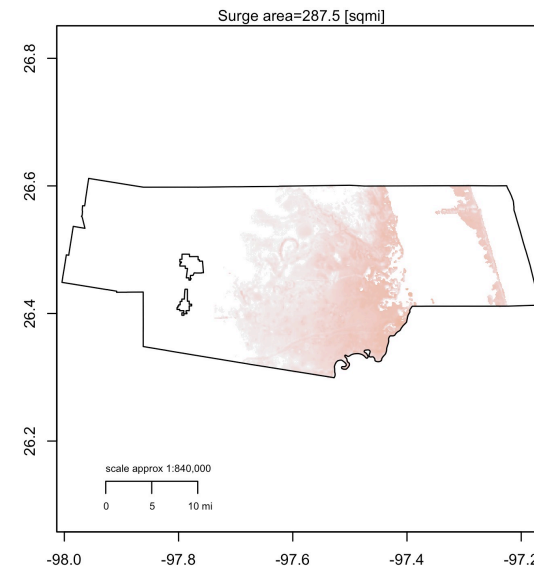
Surge cat. 2, county 48489 (no places)



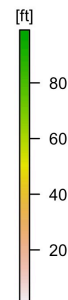
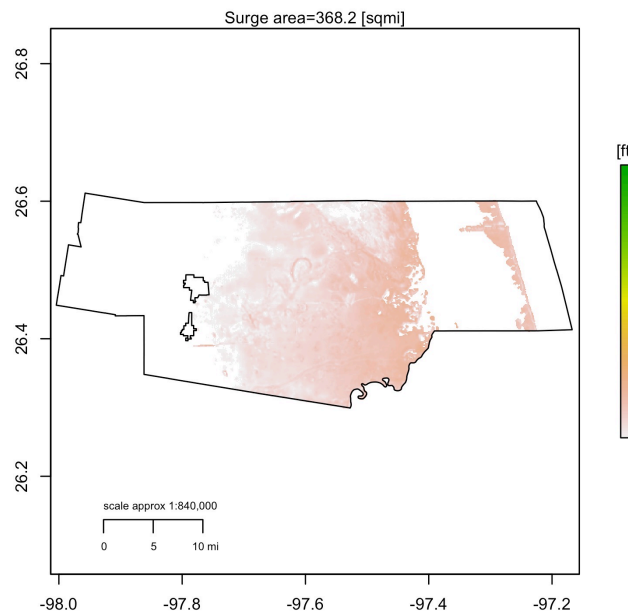
Surge cat. 3, county 48489 (no places)



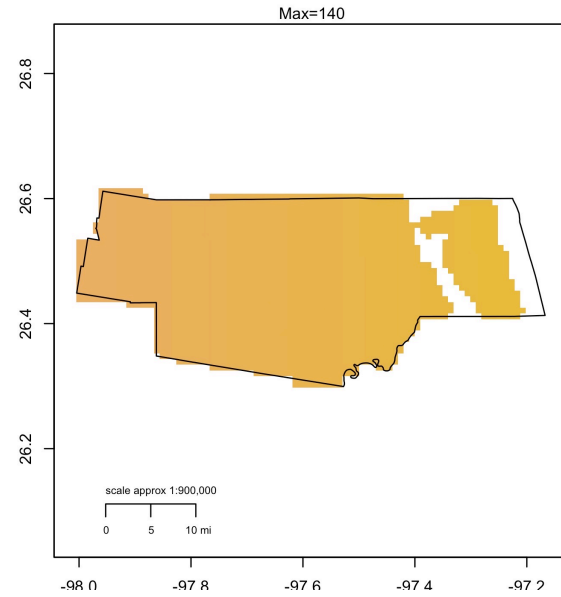
Surge cat. 4, county 48489 (no places)



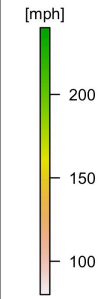
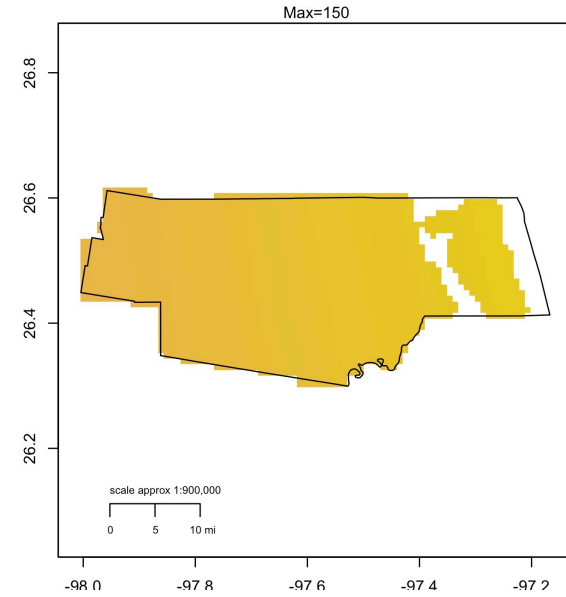
Surge cat. 5, county 48489 (no places)



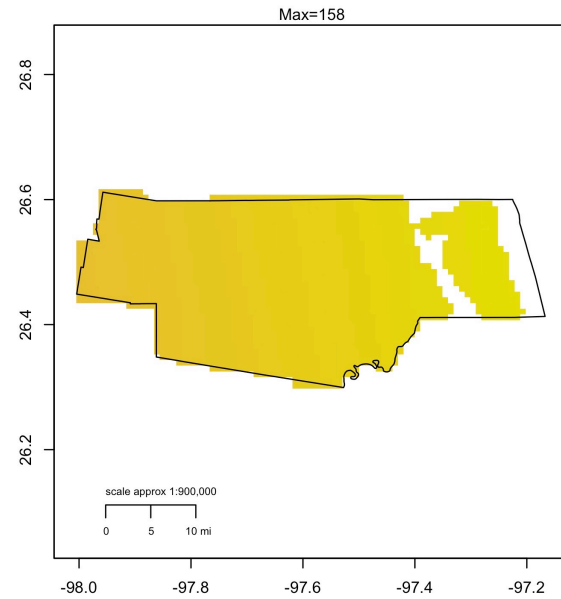
Wind risk cat. I, county 48489



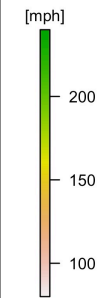
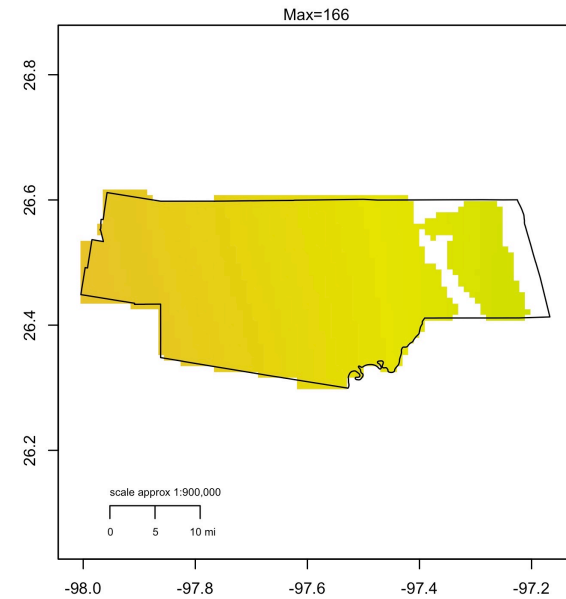
Wind risk cat. II, county 48489



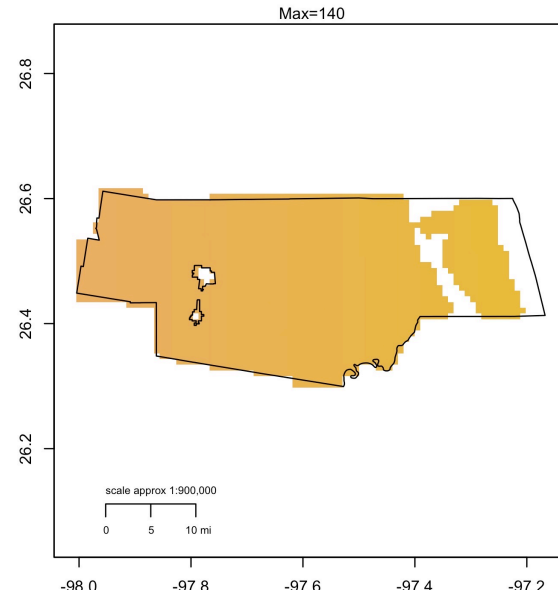
Wind risk cat. III, county 48489



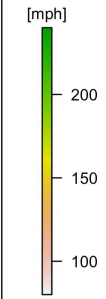
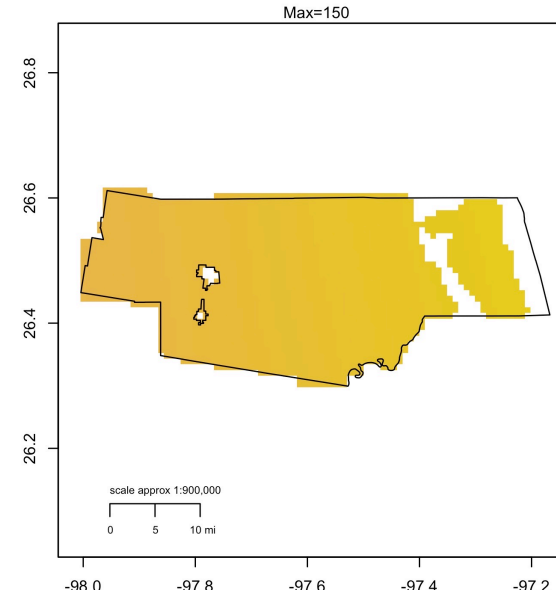
Wind risk cat. IV, county 48489



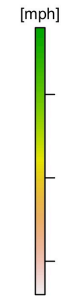
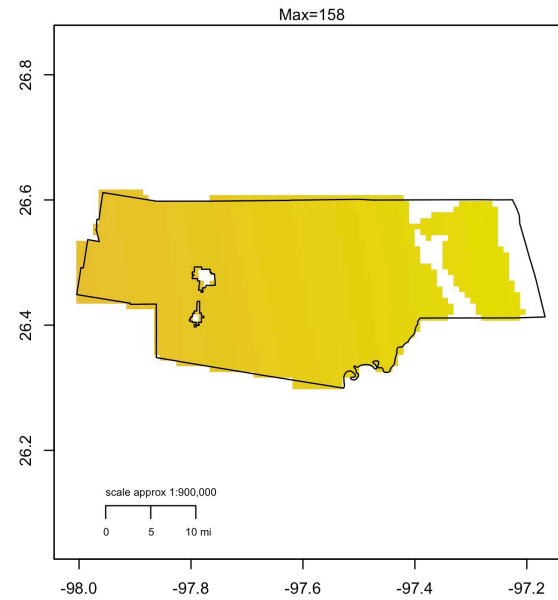
Wind risk cat. I, county 48489 (no places)



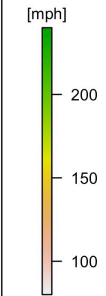
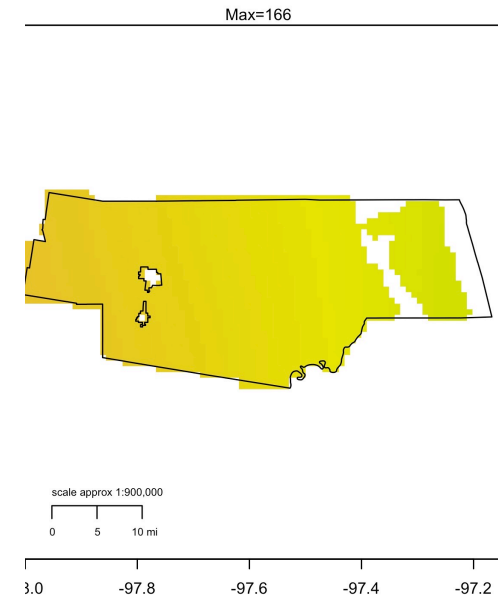
Wind risk cat. II, county 48489 (no places)



Wind risk cat. III, county 48489 (no places)

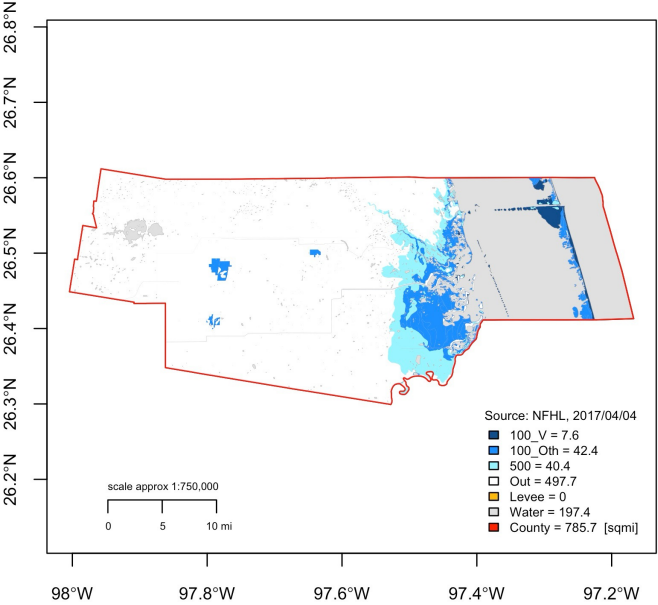


Wind risk cat. IV, county 48489 (no places)

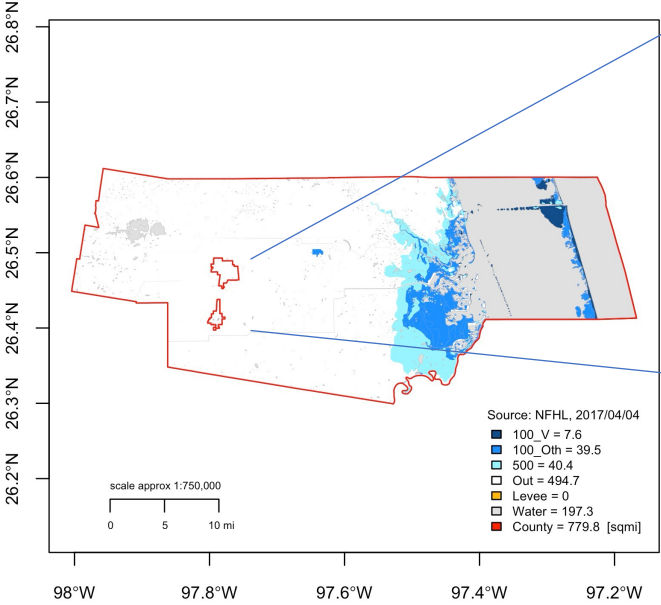


Example: surge & wind

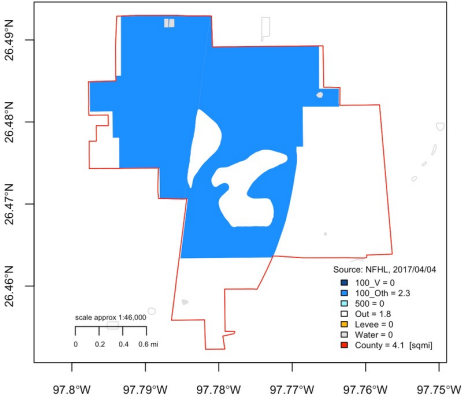
Floodplain, county 48489



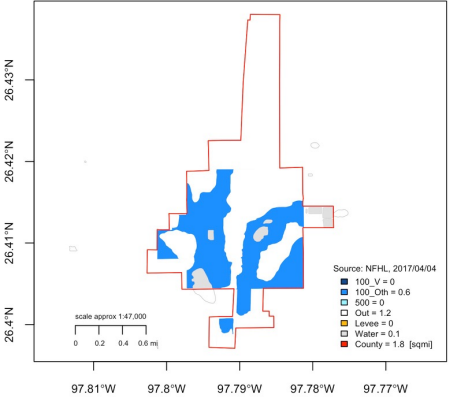
Floodplain, county 48489 (no places)



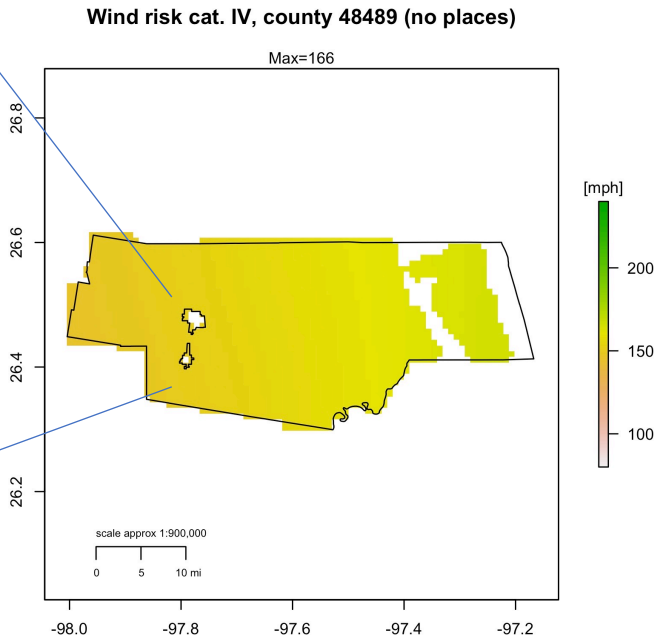
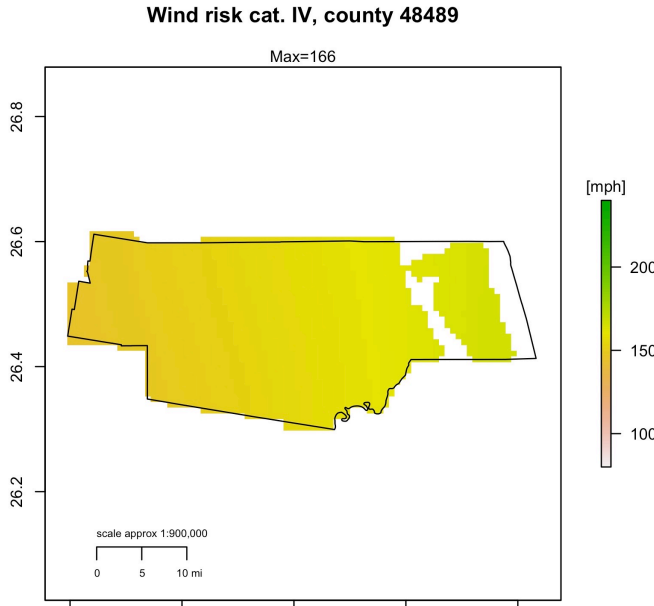
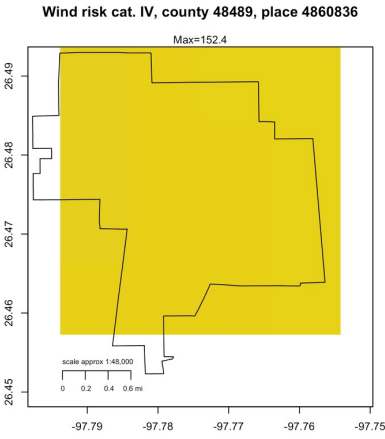
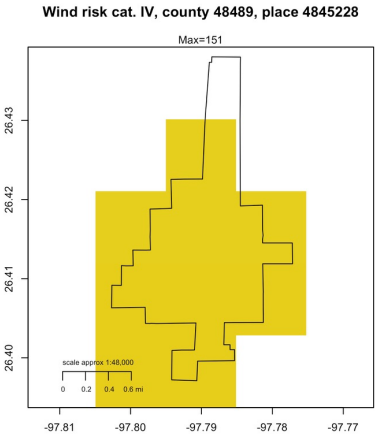
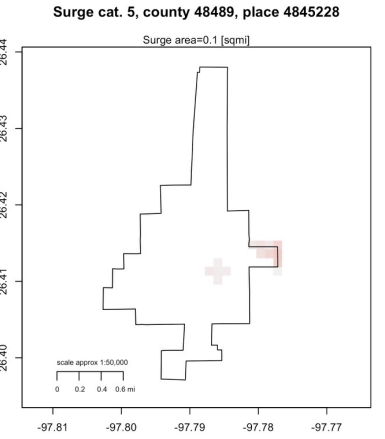
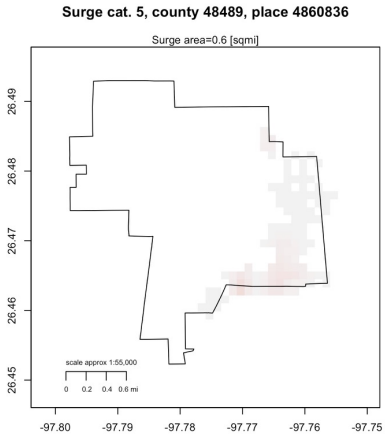
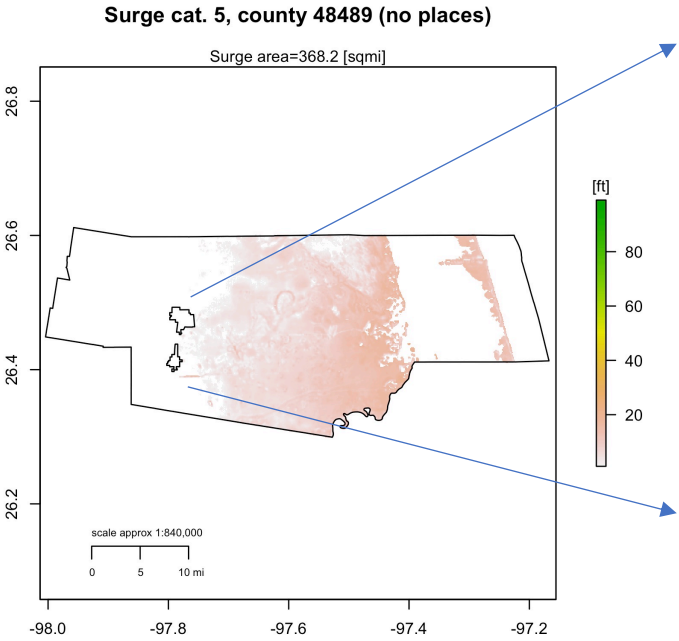
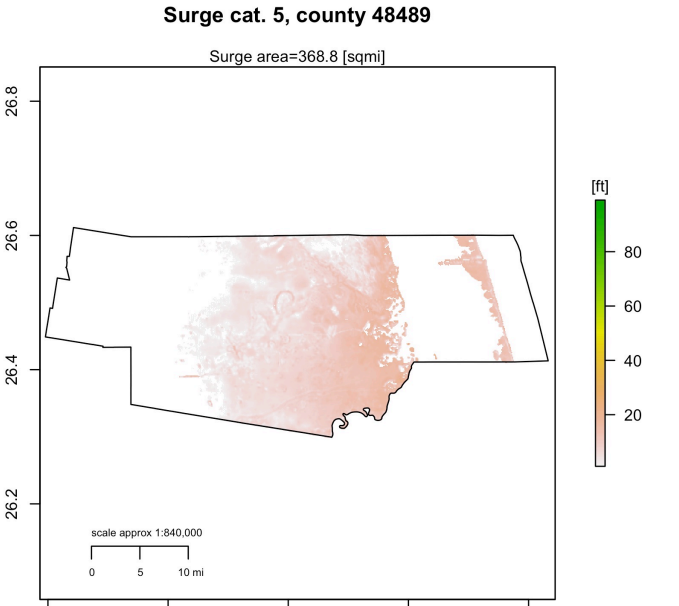
Floodplain, county 48489, place 4860836



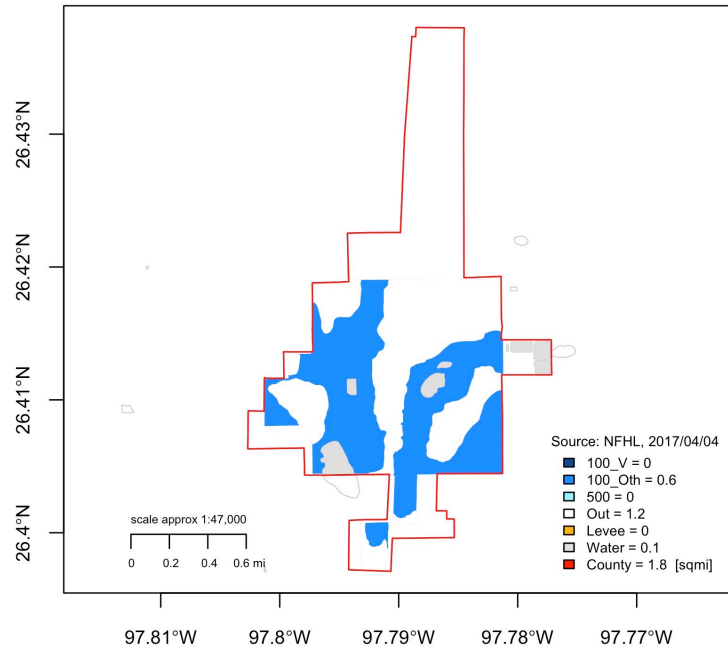
Floodplain, county 48489, place 4845228



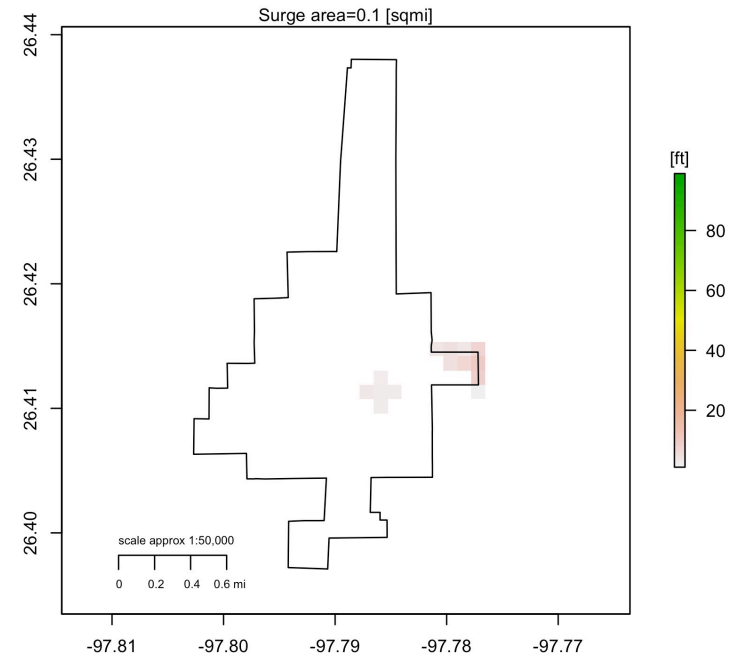
Example: surge & wind



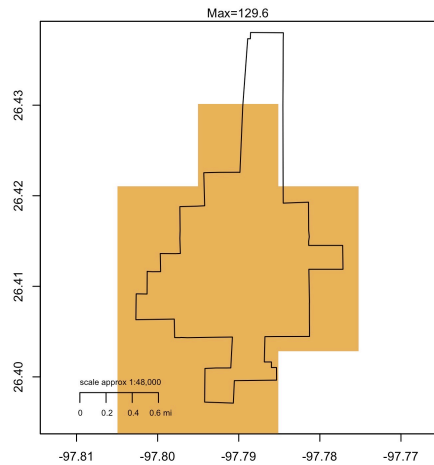
Floodplain, county 48489, place 4845228



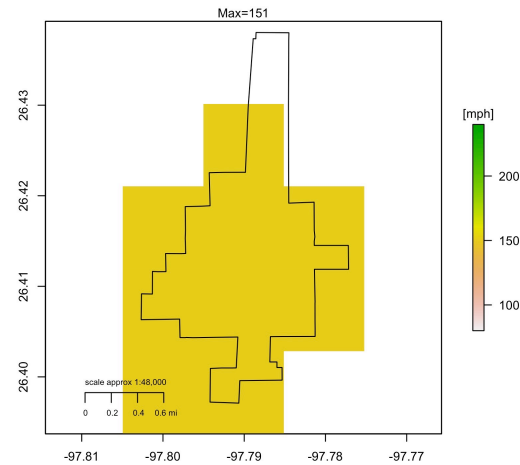
Surge cat. 5, county 48489, place 4845228



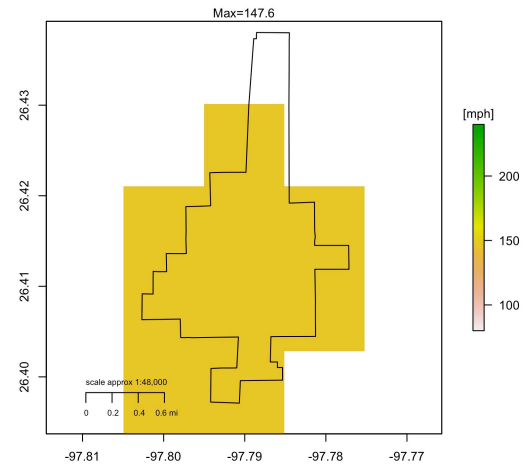
Wind risk cat. I, county 48489, place 4845228



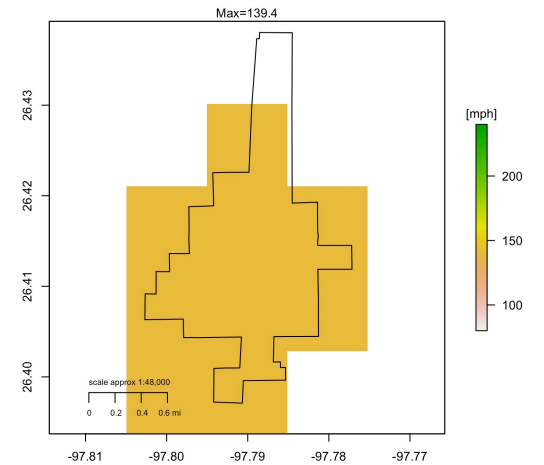
Wind risk cat. IV, county 48489, place 4845228



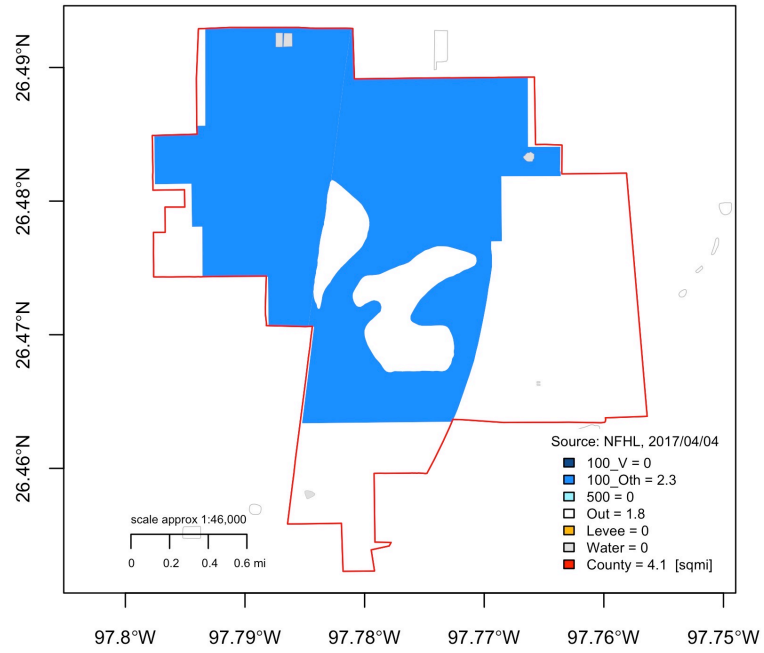
Wind risk cat. III, county 48489, place 4845228



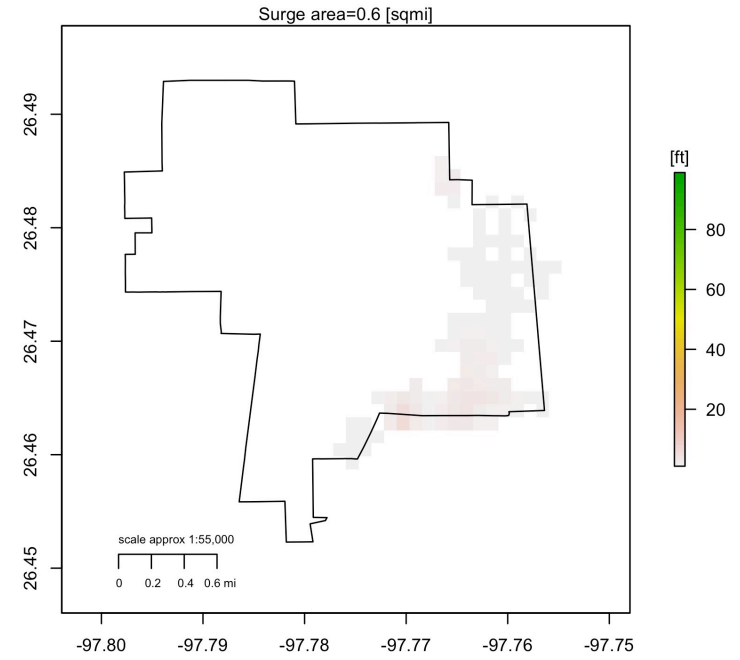
Wind risk cat. II, county 48489, place 4845228



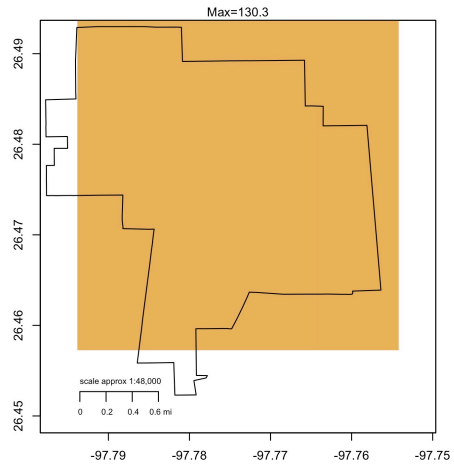
Floodplain, county 48489, place 4860836



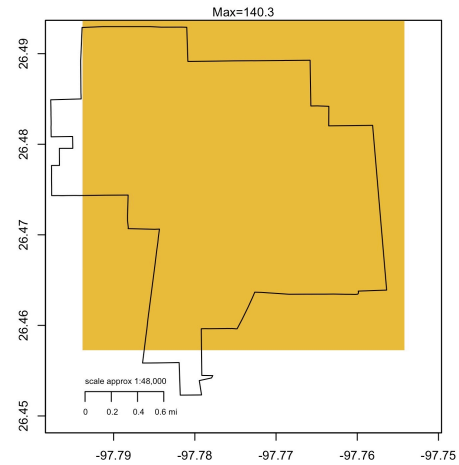
Surge cat. 5, county 48489, place 4860836



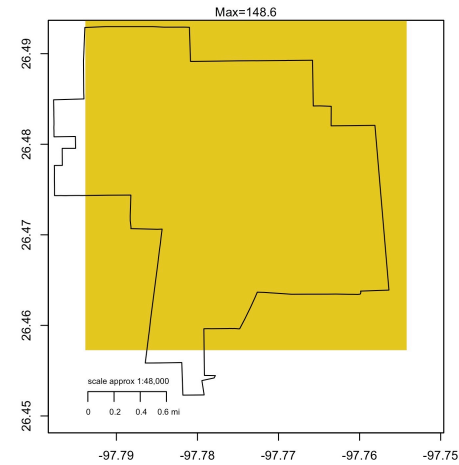
Wind risk cat. I, county 48489, place 4860836



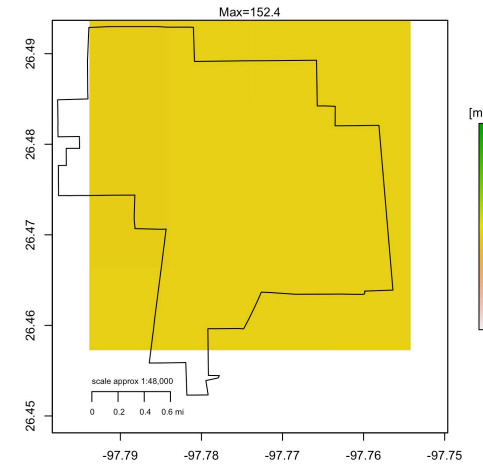
Wind risk cat. II, county 48489, place 4860836



Wind risk cat. III, county 48489, place 4860836



Wind risk cat. IV, county 48489, place 4860836

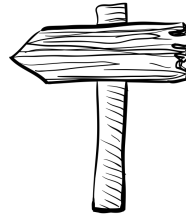
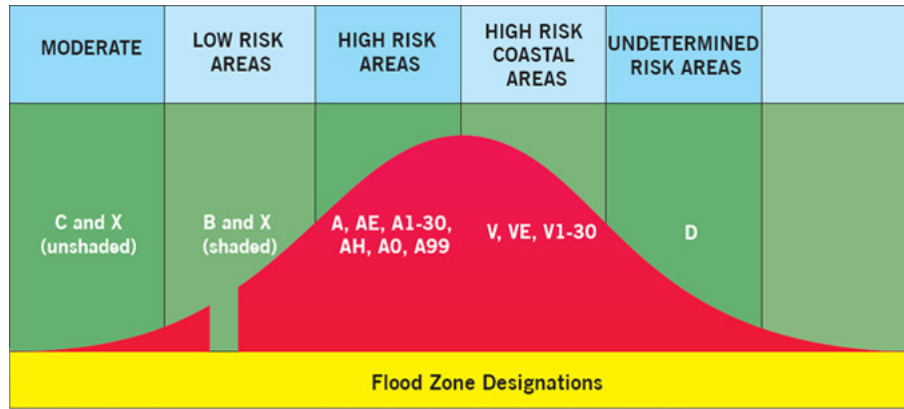


Results: partial results for Texas

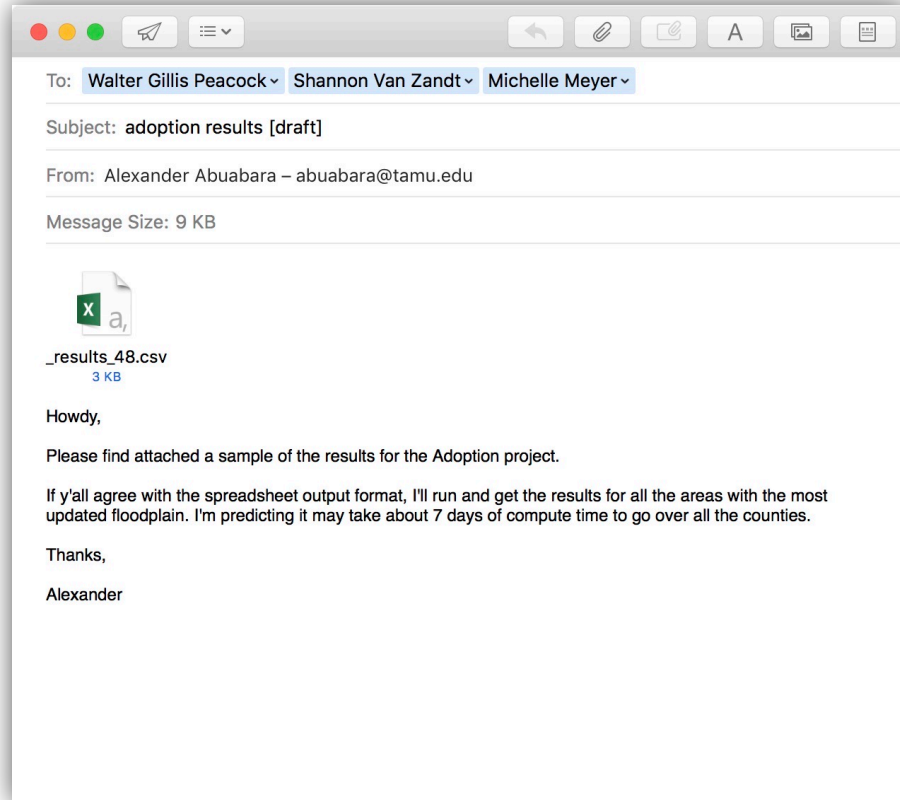
flood										surge										wind				
	county	type	no_places	no_water	m_file	Size	Source	Last_Update	Area_Total	Area_Water	Area_Levee	Area_100Yr_V	Area_100Yr_Oth	Area_500Yr	Area_Out	Area_Surge_Cat.1	Area_Surge_Cat.2	Area_Surge_Cat.3	Area_Surge_Cat.4	Area_Surge_Cat.5	Max_Wind_cat.I	Max_Wind_cat.II	Max_Wind_cat.III	Max_Wind_cat.IV
1	48167	county	NO	YES	YES	7.1 Mb	Q3	< 2005	874.9	495.5	0	0	173.2	53.3	150.7	175.8	239.3	299.9	340.2	374	140	150	160	167.5
2	48167	county	YES	YES	YES	1.8 Mb	Q3	< 2005	296.3	197.2	0	0	57.5	3.5	37.2	56.8	66.8	79.6	90.1	99.7	140	149.8	159.5	166.6
3	48489	county	NO	YES	YES	6.1 Mb	NFHL	4/4/17	785.7	197.4	0	7.6	42.4	40.4	497.7	27.3	64.3	165.6	287.5	368.8	140	150	158	166
4	48489	county	YES	YES	YES	6.1 Mb	NFHL	4/4/17	779.8	197.3	0	7.6	39.5	40.4	494.7	27.3	64.3	165.6	287.5	368.2	140	150	158	166
5	48167	place	NO	YES	YES	73.6 Kb	Q3	< 2005	2.7	0	0	0	0	0.1	2.7	0.2	0.7	3	3	3	135.2	144.9	154	161.7
6	48167	place	NO	YES	YES	533.6 Kb	Q3	< 2005	10.3	0.4	0	0	2.2	2	5.7	1.5	4.7	10.5	11	11	135.8	145.2	154.5	162.2
7	48167	place	NO	YES	YES	454 Kb	Q3	< 2005	20.8	0.1	0	0	2.2	0.6	13.3	0.5	1.5	3.7	7.6	12.4	134.3	142.7	152.3	160.1
8	48167	place	NO	YES	YES	1.1 Mb	Q3	< 2005	209.6	168.4	0	0	37.9	1.5	1.1	30.3	42.7	45	45.6	45.7	140	150	160	167.5
9	48167	place	NO	YES	YES	1.6 Mb	Q3	< 2005	92.2	31.7	0	0	42.5	9	9	34.9	53.4	63.1	64.5	64.8	139.3	149	158.7	166.1
10	48167	place	NO	YES	YES	455.3 Kb	Q3	< 2005	14.3	0.3	0	0	3.5	3.6	6.8	6.8	7.9	13.1	14.2	14.9	138	147.7	157.2	164.8
11	48167	place	NO	YES	YES	1 Mb	Q3	< 2005	53	1.7	0	0	7.2	4.5	37.5	2.4	7.6	23.6	37.5	46.8	135.5	144.5	153.9	161.7
12	48167	place	NO	YES	YES	92.6 Kb	Q3	< 2005	5.1	0.3	0	0	3.2	1	0.5	2	4.7	5.2	5.2	5.2	135.9	145.8	154.9	162.4
13	48167	place	NO	YES	YES	400.8 Kb	Q3	< 2005	17.6	0.2	0	0	0.6	0.8	16	0	0.2	2.7	6.5	14.3	137	146	155.8	163.3
14	48167	place	NO	YES	YES	864.8 Kb	Q3	< 2005	185.8	95.2	0	0	16.4	26.7	20.8	42.9	53.6	59.9	68.1	72.1	138.8	148.7	158.2	165.8
15	48489	place	NO	YES	YES	131.3 Kb	NFHL	4/4/17	1.8	0.1	0	0	0.6	0	1.2	0	0	0	0	0.1	129.6	139.4	147.6	151
16	48489	place	NO	YES	YES	82.2 Kb	NFHL	4/4/17	4.1	0	0	0	2.3	0	1.8	0	0	0	0	0.6	130.3	140.3	148.6	152.4

Suggestions ...

- Incorporate LandScan
 - Analyze the populational distribution using **LandScan**
- Reclassify the flood in terms of risk (my research does it partially):



sample results



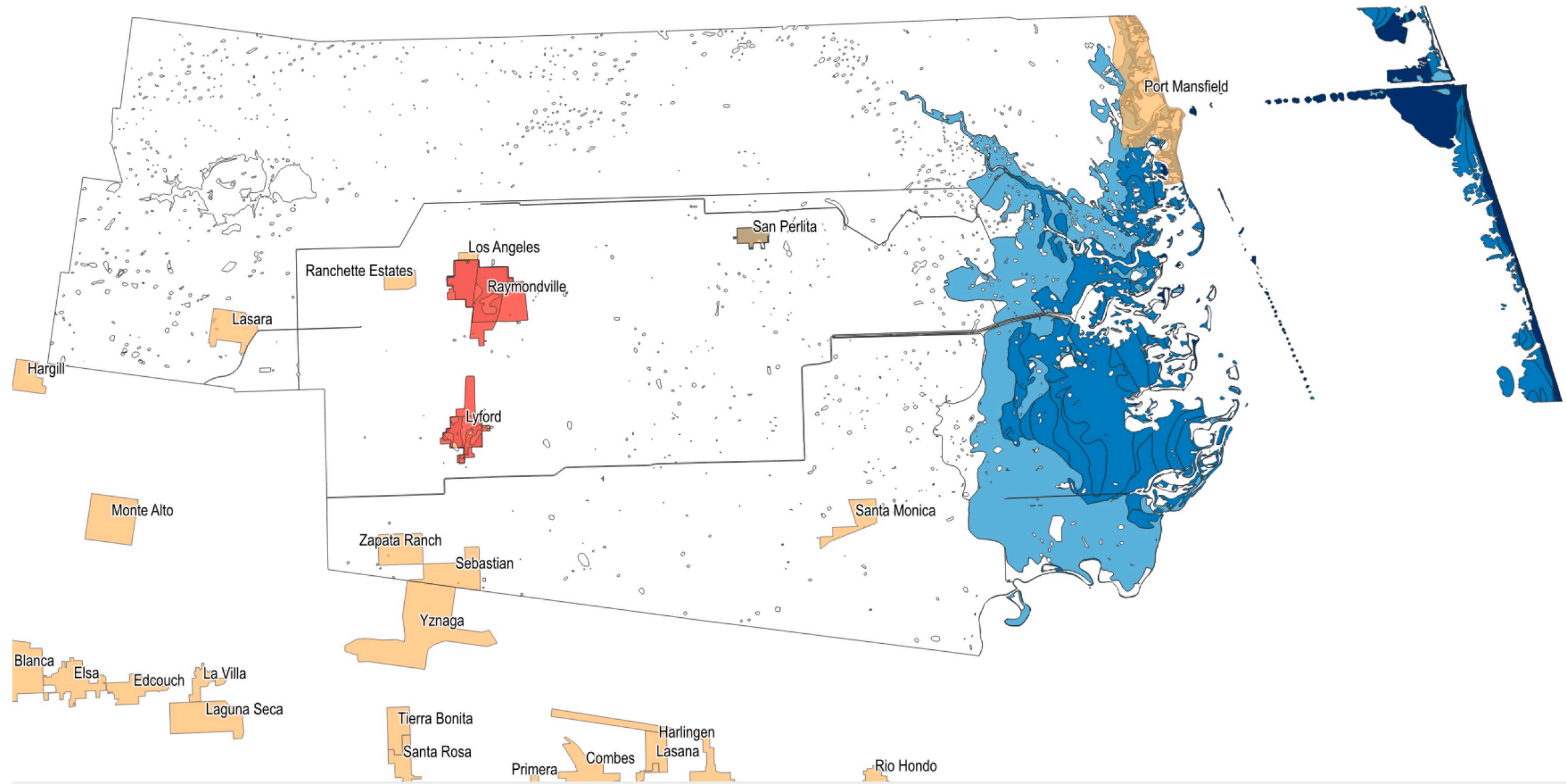
questions



- ☐ In the **county-places**, I'm subtracting just the places that **ARE** present in the *m_file* as a place, OK?

(CHECK NEXT SLIDE)

- ☐ Option: subtract all the places
- ☐ It would be good to have someone else to verify the data manually for a couple of counties, like Fayola did.
- ☐ Recommendations?



Appendix 1: buffer with width 0

- Shapefile are non-topological data formats -- they don't encode the spatial relationships between adjacent polygons and intersecting polylines.
- In order to perform topologically consistent, after each geoprocessing calculation the result need to be buffered with 0.

Appendix 2: {Choosing the Right Map Projection}

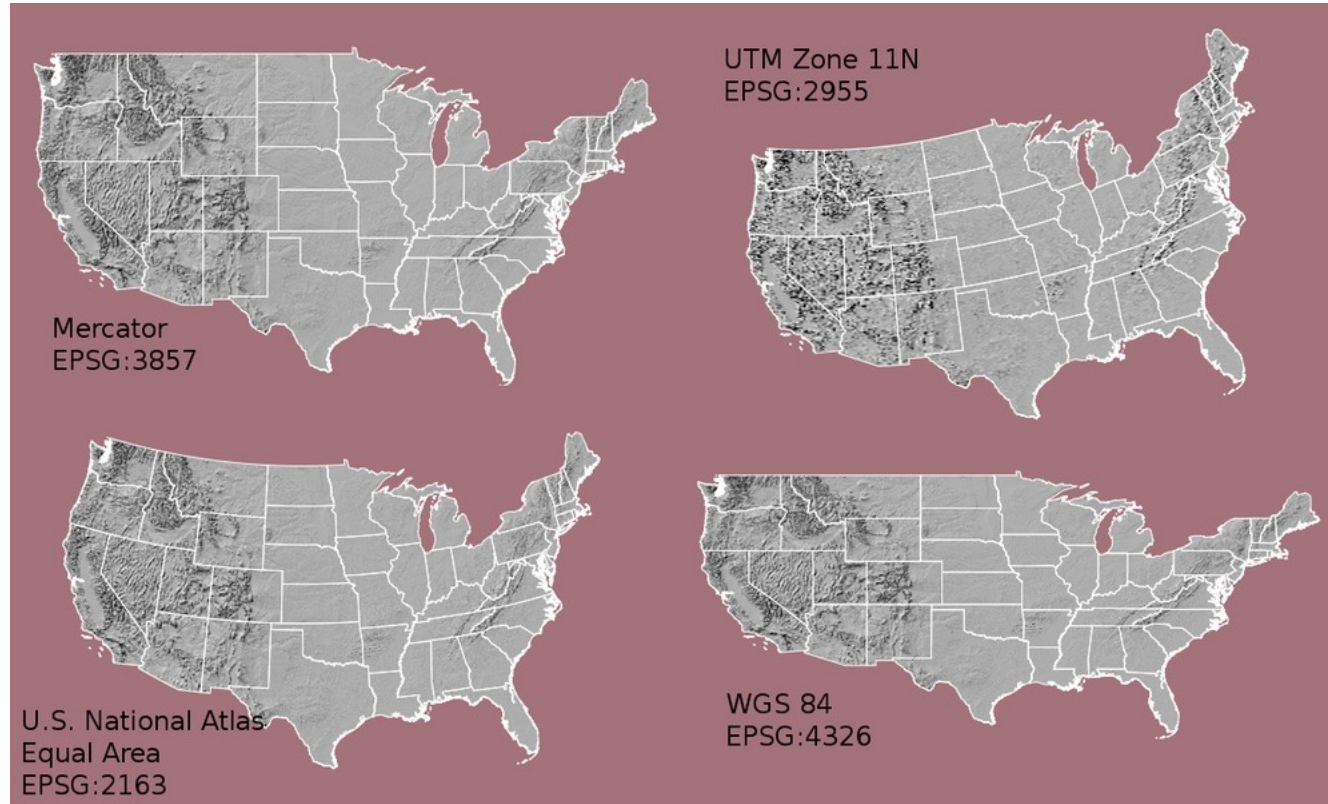
Choosing the Right Map Projection

<https://source.opennews.org/articles/choosing-right-map-projection/>

- If you're already familiar with projections and how they work (and often don't work), jump this part
- Mapping large areas: when you're mapping a large area, like the continental United States, the first concern is making sure that the projection you use is going to represent the entire map area reasonably well.
- Luckily, mapping agencies everywhere have often already flagged a good projection for whatever country you're interested in, so your first stop is to look for that.

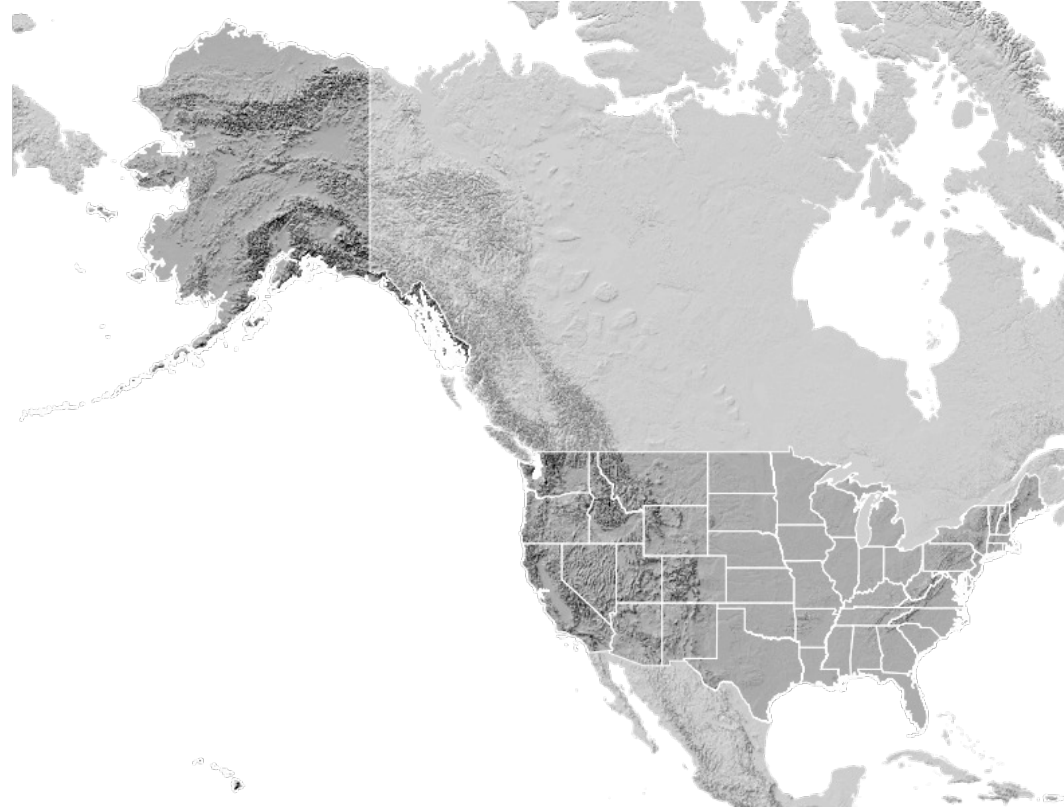
The USA four ways

<https://source.opennews.org/articles/choosing-right-map-projection/>



The United States with Alaska and Hawaii in Mercator—a bad map

<https://source.opennews.org/articles/choosing-right-map-projection/>



Choosing the Right Map Projection

<https://source.opennews.org/articles/choosing-right-map-projection/>

- The **U.S. National Atlas Equal Area** projection (**EPSG:2163**) is heavily optimized for the characteristics of the United States: wider east to west than north to south.

